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# PROVISIONAL APPLICATION FOR PATENT COVER SHEET – Page 1 of 2

This is a request for filing a PROVISIONAL APPLICATION FOR PATENT under 37 CFR 1.53(c).

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| INVENTOR(S) | | |
|---|---|---|
| Given Name (first and middle [if any]) | Family Name or Surname | Residence (City and either State or Foreign Country) |
| Christopher F. | Milam | Austin, Texas, United States |
| | | |
| | | |
| | | |
| | | |

Additional inventors are being named on the _____ separately numbered sheets attached hereto.

**TITLE OF THE INVENTION (500 characters max):**

Credit-Budgeted Maintenance and Verified Succession for Finite-Register Systems

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**ENCLOSED APPLICATION PARTS (*check all that apply*)**

☐ Application Data Sheet. See 37 CFR 1.76.

☐ CD(s), Number of CDs _____

☑ Drawing(s) *Number of Sheets* 11

☐ Other (specify) _____

☑ Specification (e.g., description of the invention) *Number of Pages* 60

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# PROVISIONAL APPLICATION FOR PATENT COVER SHEET – Page 2 of 2

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# PROVISIONAL PATENT APPLICATION

## Credit-Budgeted Maintenance and Verified Succession for Finite-Register Systems

**Inventor (proposed):** Christopher F. Milam

**Filing type:** U.S. Provisional Application under 35 U.S.C. §111(b)

The disclosure provides finite-register controllers and transfer protocols configured with the control permissions, atomic operations and resource reservations specified below. Numeric examples describe prospective operating configurations.

*The embodiments and operating examples below describe prospective configurations.*

### CROSS-REFERENCE TO RELATED APPLICATIONS

This disclosure belongs to the applicant's coordinated provisional series concerning controlled information processing and accounted operations. Provisional H's computing architecture and Provisional O's finite-operation accounting are portfolio context only. No priority claim is made, and no essential description is incorporated from an unpublished companion. The mechanisms, terms and calculation rules used here are stated in this specification.

### FIELD

The invention relates to finite-capacity information services, maintenance admission controllers, storage and replicated-state-machine scheduling, finite-horizon resource provisioning, controlled retirement and succession, exclusive-custody transfer interfaces, and upkeep of the controllers executing those interfaces. Secondary applications concern planning maintenance of finite inventories of engineered assets. Further embodiments concern bioprocess assets, cryopreserved stock, biological-sample custody, externally regulated culture systems and laboratory automation.

### BACKGROUND

1. A bounded information service accumulates obligations. A log entry may occupy a retention slot until a named release authority permits discharge to a qualified archive. A service may therefore stop for at least two different reasons: no capacity for the next posting, or insufficient accessible resources to perform the next required operation. A large nominal budget does not cure a missing release, late resource delivery, unavailable destination or conflicting custody claim.
2. Moving a checkpoint to another server does not necessarily release its pending obligations, move its service rights or pay for verification. Conversely, retiring a complete module and assigning a continuing service role to a supplied successor can change which obligations the active module must answer. These operations require different contracts and accounting; changing an identifier alone accomplishes neither.

3. A migration descriptor can arrive late, corrupted or duplicated while the assets it describes remain in exclusive escrow. Treating the descriptor itself as a transferable credit or authorization permits inconsistent balances or repeated assignment. The interface must compare the proposal with actual authoritative state, reserve the intended destination, recheck terms at commitment, retain rejected attempts, and distinguish assignment from final readiness.
4. The controllers that verify and commit transactions also consume finite resources. A maintenance budget therefore accounts for controller upkeep, session setup, reference replacement and retained state. The system preserves spent-resource records and charges each subsequent opening for the resources and initialization operations that opening requires.
5. The disclosure uses finite-register credit accounting from the POAMS framework to configure maintenance, financing, succession and transfer-interface operations. Credit units, pricing menus, arrival profiles and permission-controlled fields are stored design inputs. The controller uses the calculation rules to allocate resources, enforce operation permissions and execute guarded state transitions. The disclosed subject matter comprises these engineering mechanisms and protocols.

## SUMMARY

A finite-register service system (1005; FIG. 1000) includes a bounded obligation store (1010), a qualified-resource registry (1020), a maintenance admission controller (1030), a horizon provisioning engine (1040), a succession engine (1050), a transfer interface (1060), and verifier, escrow and commit cores (1070–1090). The registry identifies actual resource holders, release rights, deadlines, destinations, spent-resource capacity and operation prefixes.

Five cooperating families are disclosed:

- **P1 — Maintenance admission.** A controller reserves and debits ordinary work, two-credit discharges and other distinct fees against a finite-fuel envelope, checks delivery/release/custody qualifications at the applicable prefix, executes admitted operations, and reports funding halt separately from overflow and qualification interruption.
- **P2 — Horizon provisioning.** An engine converts a requested horizon into ordinary and maintenance reservations using effective spare capacity; it rejects money-only feasibility when the required stock, permissions or timely routes are missing. An optional fixed-model survival calculation accompanies the actual allocation.
- **P3 — Verified succession.** An engine selects a typed transfer law, retires a module only under standing authority, assigns exclusive unexpired rights to an already eligible successor, preserves required records and residual clocks, and inhibits a subsequent session without a paid and qualified initialization transition. A restricted status-snapshot format is provided for the declared status-only family.
- **P4 — Transfer boundary.** A canonical descriptor identifies actual escrowed assets. Full comparison, guarded reservations and once-only assignment prevent the descriptor from becoming

an asset. Attempt, verification and configuration reserves are separate, and obstruction records prevent futile price escalation from being reported as completion.

- **P5 — Budgeted guardians.** Finite verifier, escrow and commit machines perform actual-field reads and permission-confined dispatch, maintain separate obligation books, and must pass their own funding, capacity and readiness checks before permitting the next transfer stage.

Biological embodiments B1–B5 apply these controllers to bioprocess maintenance, lineage succession from preserved stock, biobank escrow, credit-controlled culture-factor delivery and guardian upkeep in biological facilities. The mechanisms enforce asset-specific acceptance rules, actual custody, immutable rights and paid initialization through the common transaction and qualification interfaces.

**B6 — Adversarial sub-asset governance.** A read-only audit subsystem monitors observable custody and posting evidence, computes exact-at-count detection power and separately certified minimum-count floors, and prices both correct and false-positive responses. A configured decision controller enforces authorized supply cutoff and retained-state quarantine, then selects containment or whole-module retirement with lawful rights transfer using qualified work/capital comparisons. The same governance loop applies to distributed ledgers, storage fleets and managed biological assets.

The principal embodiment is a permissioned replicated storage service with bounded retention registers and a designated ordered executor. It supports log pruning, archival succession and resource-limited maintenance while retaining the authoritative custody record. Obligation slots, retention permissions, resource reservations and committed prefixes are named and checked directly. Secondary fleet-planning embodiments specify separate mappings for asset deterioration, restoration and the corresponding operation prices.

## BRIEF DESCRIPTION OF THE DRAWINGS

Eleven figures use the H-series convention of one thousand-series reference-numeral family per figure. The accompanying drawing set supplies eleven sheets.

- **FIG. 1000** — Finite-register service architecture, qualified-resource registry, five controller families, and separation of target, interface and guardian accounts.
- **FIG. 2000** — Maintenance admission state diagram, ordered guards, paid operation prefix, and distinct qualification, funding and overflow exits.
- **FIG. 3000** — Horizon frontier chart, effective spare capacity, integer allocation, and a qualification mask excluding nominally funded but unauthorized plans.
- **FIG. 4000** — Succession modes and dominance map: four modes showing what moves, what remains, common comparison cuts and dimensions of cost, service survival and residual stock.
- **FIG. 5000** — Minimal restricted status-snapshot encoding and its boundary against payload, cause, policy and audit records.

- **FIG. 6000** — Capsule/escrow protocol sequence with full verification, rejected slots, padded common cut, assignment and later readiness receipt.
- **FIG. 7000** — Verifier, escrow and commit core state machines, separately budgeted upkeep and finite recipient/spent headroom.
- **FIG. 8000** — Completed-session interlock, retained state, cumulative upkeep and inhibited re-entry without separately priced and qualified reinitialization.
- **FIG. 9000** — Bioprocess admission, factor delivery and guardian control.
- **FIG. 10000** — Biological succession and biobank custody sequence.
- **FIG. 11000** — Adversarial sub-asset governance and retained-state quarantine.

## DEFINITIONS

The following definitions govern the disclosed operating domain and claims.

1. **Register, slot and obligation.** A register is a finite inventory of B addressable service slots. An occupied slot retains a named pending obligation and any payload or references required by the declared service contract. An archive outside that active inventory is still retained stock, not deleted history.
2. **Service role; root.** A role is a specified continuing service contract. A root identifies its current endpoint. Continuation of the role by a successor is not continued operation of the retired root.
3. **Credit and work account.** A credit is a discrete, exclusively accounted authorization/resource unit under a stored operation-price menu. It is not automatically currency or physical energy. An ordinary word costs one credit; a selected single-slot discharge costs two additional maintenance credits. Other prices are separately declared. Actual spendable, locked, inaccessible and spent balances are distinguished.
4. **Finite-fuel profile.** The selected operation family admits no spent-work reclamation or unrecorded outside input. $V_0$ is opening spendable work stock; $T_0$ and T_min are the opening and minimum retained non-work tariff of a specified convertible account; ε_W is its positive tariff per work credit. The bound $V_0+(T_0-T_min)/\varepsilon_W$ is an upper envelope, not an implemented conversion. If no converter is supplied, only actual available $V_0$ funds execution. $T_0$ is not a time horizon.
5. **H, D and F.** In the whole-boundary budget H counts all charged ordinary words, D counts completed or reserved two-credit single-slot discharges, and F counts other distinct work fees. Preparation and guardian ordinary words must either be included in H or explicitly included once in a separate subledger. Elsewhere H denotes the specified target-service horizon, and H_total is used for the whole-boundary count.
6. **Qualification.** A resource is qualified only when the actual amount is accessible on its admitted delivery route at the required stage, the addressed obligation has a valid release, the receiver is distinct and ready, custody is exclusive, deadlines and eligibility hold, and blank, code, retained-

record and spent headroom are available. A nominal balance or average delivery rate alone is insufficient.

7. **Funding halt F_h; overflow O_h; qualification interruption Q_h.** F_h means the required next payment cannot be funded. O_h means an admitted arrival has no available service slot. Q_h means a required nonmonetary guard fails. A paused device retains its recorded state for authorized resumption. Fee F is not the label F_h.
8. **Qualified fixed-horizon profile J.** $B \geq 1$; opening occupancy $z \leq B$; $c = B - z$; one binary arrival per funded normal word; maintenance before the unrevealed arrival; a local maintenance purse admitting at least two credits; enough correctly addressed, timely releases and independent blank recipients; finite source/code/spent resources; no recipient reuse; and allocation fixed before the opening cut. For probabilistic calculations, arrivals are IID Bernoulli(a), $0<a<1$. Robust all-word provisioning does not require IID probabilities.
9. **Effective spare capacity.** If M maintenance credits are accessible, Y distinct usable blanks and R universally applicable release uses are qualified, $m = \min(\lfloor M/2 \rfloor, Y, R)$, further capped by any service-opportunity limit, and $\ell = c + m$. R is not a count of unrelated authorizations.
10. **Right token.** A unique serial with issuer, holder, permitted assignees, scope, remaining uses, immutable absolute expiry, expiry clock domain and the authoritative issuer-approved source of that clock. Assignment changes the holder only as authorized; it does not duplicate uses, confer new eligibility or extend expiry.
11. **Capsule.** A canonically encoded proposal describing a particular reserved asset bundle, session, attempt and target. The capsule is not the bundle, a credit, a right or a completion receipt.
12. **Common cut.** A prospectively fixed comparison stage reached with the same paid preparation duration in both compared branches. Faster completion is padded with paid IDLE words where specified. Deadlines and residual clocks are evaluated at the actual advanced cut; an expiry-sensitive operation also requires the authoritative current clock check in E1/E6, and a planned cut cannot substitute for that current value.
13. **FC-1 field-read/dispatch primitive.** The control substrate supplies reads of named finite actual-state fields into supplied blank pointers and finite-program dispatch of the specified operations from those readings while retaining displaced labels. Every operand path enforces the stored issuer/lock permissions. The admitted fault family affects proposal carriers and excludes these reads, tables and permission wires. The embodiment requires these substrate capabilities as control primitives.
14. **Atomic assignment.** One guarded operation updates the listed custody, purse, title and role coordinates together with its durable completed-event marker and next-prefix value inside the ordered transactional state-machine boundary specified in E9. It is an expressly supplied primitive. A multiword transfer can stop after a completed proper prefix; earlier moves remain committed and final public readiness requires its separate receipt.
15. **Configuration and setup.** Configuration arranges supplied finite factors under $K_0 + \kappa N$. Session setup initializes actual escrow, reference and clear prefixes under a separately stated price J.

Neither term produces raw materials or manufactures credit.

16. **Session re-entry receipt.** A record binding the prior completed state, proposed new initialized state, retained displaced records, actual payments, stock, authorities and exposure schedule. Its checking is an admission requirement. A receipt does not construct an absent transition.

## DETAILED DESCRIPTION

### E1 — Lead storage and replicated-state-machine embodiment

The system 1005 operates a bounded retention service. A slot 1010 contains an obligation identifier, retention contract, current payload reference and occupied flag. A complete physical or digital storage roster distinguishes active service slots, archive recipients, proposal buffers, immutable receipts and controller work registers. Extra metadata memory does not increase service capacity B. An operation table identifies which queries remain promised after pruning; a release cannot discard a payload that the active service must still answer without a separately qualified response path.

One or more processors are configured to execute the controller against an authoritative finite registry 1020. A permissioned replicated deployment may use an existing ordered transaction service as the designated executor. All resource mutation and release interfaces route through that executor's permission checks; client-side cached descriptors cannot authorize spending. Replica delivery, ordering and persistence must meet the selected implementation's availability and fault assumptions. The deployment uses the selected transaction service's consensus and durability facilities, with trusted-executor corruption outside the specified fault domain. The chosen permission and transaction substrate supplies the required FC-1 and atomic-assignment capabilities.

Every right identifies its expiry clock domain and the authoritative source of that clock. In the lead storage implementation, a wall-clock lease uses issuer-authorized current time, including elapsed suspension time; a logical-epoch right uses the issuer's authoritative epoch. A controller's paused local step index is not substituted for either source. Immediately before each expiry-sensitive assignment or use, the executor obtains the authoritative current value and checks the immutable term against that value. If time cannot be established, the operation is qualification-inhibited. A planned comparison cut is an accounting reservation, not evidence that a late commit remains timely. An interruption beyond the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate; resumption preserves completed mutations but must recheck remaining rights, headroom, exposure, funding and actual time before executing the outstanding suffix. Missing qualification is not cured by charging more credits. Assignment already completed before the interruption is retained without reversal or repetition when the final receipt is late.

A work credit can represent a prepaid scheduler unit whose operation class, source reservation and recipient are recorded. The reference menu uses ordinary 1 and maintenance 2 in scheduler-credit units. Mappings to device energy, monetary fees or another implementation's processing costs are specified

separately. A changed menu requires a changed controller certificate and recomputed bounds. Opening accounts may set $T_0$=T_min, making the finite-fuel envelope exactly $V_0$ without invoking a tariff converter.

The controller freezes a configuration tuple `(profile_version, operation_family, horizon, boundary_ID, query_family, stage_order, price_menu, stock_roster, permission_roster, fault_family)`. It retains the exact values used for an admission decision. A change of query family, fault boundary, release route or price is a new declaration; a stale certificate cannot authorize it. Actual resource operations and retained records, rather than an estimate alone, complete the mechanism.

Each authoritative fine-stage command uses the immutable event key (profile_version, session, attempt, stage, operation_index). Resource mutation, its spent and retained-record entries, the completed-event marker and next prefix commit as one durable atomic update. A completed key returns its stored outcome without re-executing the mutation. E9 specifies the recovery and external-side-effect boundary; the key controls atomic execution and recovery.

**E2 — P1 maintenance admission and interruption discrimination**

FIG. 2000 identifies the stage guards 2030/2040, paid action 2050, retained prefix 2080 and suffix-only resume 2090.

At an opening cut, the controller records actual resources and any separately specified converter. It calculates the finite-fuel upper envelope and checks a proposed schedule against:

$$H_total + 2D + F \le A_0,$$
$$A_0 = V_0 + (T_0 - T_min)/\varepsilon_W. \qquad (1)$$

All quantities are expressed in the declared credit/tariff units; integer reservations cannot round a fractional envelope upward. Equation (1) is necessary in the finite-fuel profile. It is not sufficient to authorize a payment. Actual conversion, delivery and rights must make each planned debit possible. A profile admitting reclamation or new outside input is not certified by (1) unchanged; actual input is receipted and the applicable account recomputed.

The maintained audit equation posts each debit to one spent recipient with a unique event identifier. Transfers between purses change ownership/access but not aggregate work. A two-credit discharge exchanges the complete addressed obligation into a distinct qualified recipient, releases one active slot and retains the old record under the declared no-recontact or service-response boundary. Canonicalization or removal of a payload distinction receives discharge credit only when it releases an active slot; otherwise its fees enter F and it receives no discharge credit. The controller never subtracts the same fee through both H_total and F.

The default ordered word is receive/ordinary payment → nonnegative top-up → maintenance → encounter → clock → receipt. A prospective admission check reserves needed stock; the controller records completion separately after execution. Actual stage guards are then checked again. At a stage with multiple failures the stored priority is Q_h before F_h; a later arrival overflow is O_h. A denied optional maintenance action is not automatically a funding halt: the funded word may continue without discharge and a subsequent arrival may overflow. The receipt records the actual first obstructing stage and retains all already executed debits and moves.

For a local maintenance purse w of capacity W, the selected top-up amount is:

```
u = min(max(0, required − w), accessible_donor, W − w). (2)
```

Valid balances satisfy $0 \le w \le W$. Invalid balances are rejected, not clipped into validity. An interrupted word retains its stage index and completed event identifiers. Resumption executes only the unpaid/unexecuted suffix. The paid resource change, spent entry, retained-record references, completion marker and next prefix belong to one atomic durable update keyed as in E9; recovery sees all of them or none, and returns a stored completed outcome without repeating the operation. A retry is a newly accounted attempt, not a replay that silently repeats a debit or asset transfer.

Funding halt and overflow counters are distinct. A request with no ordinary credit and an empty store is a funding halt, not a full register. If the service profile asks for uninterrupted continuation, that halt is a failed prefix even if the device may later resume. Qualified continuing external supply can prevent overflow over a declared horizon, but an average supply promise without timely release and delivery cannot be substituted.

**E3 — P2 horizon provisioning and fixed-profile survival**

FIG. 3000 shows the fixed-horizon frontier and qualification mask 3010; the mask is applied before issuing the robust certificate.

For profile J the engine first reserves H ordinary credits. If E<H it returns no funded H-word plan. Otherwise it assigns maintenance pairs and verifies corresponding releases, blanks and timing. With sufficient qualifications:

```
m = floor((E−H)/2),        ℓ = c+m,
F_ℓ(H;a) = Σ[j=0..min(ℓ,H)] binom(H,j)a^j(1−a)^(H−j),
S∗_H(E) = 0                              if E<H,
S∗_H(E) = F_(c+floor((E−H)/2))(H;a)    if E≥H.        (3)
```

When stock caps bind, m is reduced as in Definition 9. The engine must not display the uncapped optimum for an unqualified plan. Equation (3) is the fixed-horizon, qualified policy-class optimum, not an optimum over arbitrary adaptive transfers of unused reserves or correlated arrivals.

For a robust H-word request the provisioner allocates:

$$E_{required} = H + 2[H-c]_+, \qquad [x]_+ = \max(x,0),$$
$$\text{required qualified services} = [H-c]_+. \qquad (4)$$

It sends the allocation and resource identifiers to the admission controller. The implemented policy discharges once when the register is full and a qualified service remains, before observing the next arrival; otherwise it waits. With at most one binary arrival per word this full-trigger policy supplies effective spare capacity c+m. It cannot use knowledge of an unrevealed future arrival to avoid the price of a pre-arrival discharge.

For a fixed credit stock E the largest robust horizon in this class is:

$$H_{rob}(E,c) = E \quad \text{if } E \le c,$$
$$H_{rob}(E,c) = \operatorname{floor}((E+2c)/3) \quad \text{otherwise.} \qquad (5)$$

The schedule reserves resources for the chosen horizon. The engine does not combine separately optimized horizons into one cohort process or infer its hazard from ratios of their separate optima. For a fixed ordinary reserve O and fixed $\ell$, overflow probability by O is $1-F_{\ell}(O;a)$; surviving paths then encounter funding halt upon the next requested word. Cause-specific receipts therefore remain meaningful even when a horizon survival value is reported.

Example: a configuration with B=3,z=2,c=1,H=4 requires E=10 and three qualified discharges for robust service. At E=8, m=2 gives $\ell$=3 and survival 15/16 for a=1/2. The controller withholds the robust certificate when a required release is missing, including when E=10. These values specify the configured credit allocation and fixed-profile probability calculation.

### E4 — P3 succession modes, rights and comparative decisions

FIG. 4000 identifies transfer modes 4010–4040 and the aligned-cut and separate-result panels 4050/4060.

The succession engine accepts a service-role contract and one typed transfer mode. Its stored mode-specific operation table maintains separate ordinary-operation, two-credit discharge and transfer-fee reservations; moving an occupied record is never credited as a discharge that releases an active slot. The original and successor modules, recipients, rights and banks are all on the opening roster or enter as receipted outside supply. It selects among the following modes without mixing their promises:

| Mode | Operation | Continuing obligations and custody |
|---|---|---|
| R — record-only | Move named complete record and residual-clock factors to an authorized endpoint | Old pending answers and burden remain; no purse is implied |
| U — rights-only fresh role | Retire the whole old module under standing authority and assign the named continuing role/right bundle to a supplied fresh root | Old payloads stay in retained old-module custody; only the separately authorized renewed contract is active |
| F — priced full transferable | Move specified records, existing purse amount, transferable rights and capacity titles | Old history and identity are not erased; transferred occupancy is not discharged |

| Mode | Operation | Continuing obligations and custody |
|---|---|---|
| custody | | |
| L — lineage re-rooting | Reassign the named source-role/root while transferring only listed existing funding | Actual conditioned source state, ancestry, dormant balances and selectors remain; no independent source is redrawn |

R/F must carry every payload distinction that the successor's promised queries can inspect, or provide a separately qualified response path. A status synopsis cannot replace those answers. A timer-sensitive transfer carries the advanced residual timer at the actual cut. L preserves the actual conditioned source state when changing the root; use of the IID arrival law in equations (3)–(5) requires its separate source-profile conditions.

Before U, the engine verifies preissued whole-module retirement authority and successor eligibility. Immediately before each expiry-sensitive assignment or use it also checks the issuer-authorized current clock in the right's declared domain under E1/E6; a missed reserved cut does not freeze external expiry. It records the old root as retired and preserves the old module as a named archive. Assignment changes the exclusive holder of the existing right; issuer, scope, uses and absolute expiry remain unchanged except for an authorized consumed use. If the next useful operation is later than the absolute expiry, assignment is refused as a qualification solution. An already issued successor-contingent grant can be exercised according to its original terms; it is not a newly minted grant. A separate issuer novation would be a different receipted operation outside the four default modes.

The comparison engine aligns preparation durations, resource rosters and future query families. At a qualified postcommit cut with capacity c_X and available service count m_X it evaluates effective capacity ℓ_X=c_X+m_X. For funded IID continuations through H, a strict survival improvement occurs precisely when min(ℓ_X,H)>min(ℓ_Y,H). It reports actual work, reserved capital, service probability, elapsed words and residual stock separately, with each comparison tied to its specified dimension.

Under fungible equal future resources, n aligned preparation words and d early discharges, the reset comparison is:

```
(c+d) + floor((E−n−2d−H)/2)
  = c + floor((E−n−H)/2).                    (6)
```

Thus clearing a register early does not itself buy a financing discount. A different result can arise from expiring release rights: exercising them while valid may supply capacity unavailable to deferred service. The controller records that rights-timing difference, not an erasure benefit. In the declared two-full-slot example, E=8,H=2 and two common preparation words with releases valid only during preparation give survival 1 after paying both early discharges, versus 1/4 when both are deferred under a=1/2.

A separate U example uses the same opening pair of capacity-two modules in both branches, one full and one empty. With two paid preparation IDLE words, assignment surcharge 1 and two future service words, U spends 5 and survives every future word. Qualified in-place deferred repair spends 6 or 8, depending

on the first arrival, and also survives every word. The comparison engine reports this service-role/work comparison for that operation menu, records the old root as retired, and accounts for leftover blank factors separately. For an integrated interface-and-guardian configuration, it adds the corresponding charges below before selecting a branch.

**E5 — Restricted minimal succession snapshot**

FIG. 5000 identifies the cap, encoder, envelope and decoder 5010–5040 and the excluded-query panel 5050. Reject branch 5060 handles unused, noncanonical or mismatched codes without constructing a live state.

This subsection applies only to the fixed qualified profile J with independent Bernoulli(a) future arrivals for a fixed $0<a<1$, no additional conditioned-source variable, the fixed full-trigger policy, and status-only observations without archive feedback; its active prediction state is (live,o,ℓ). Here H is the remaining integer horizon, o is actually qualified ordinary words remaining, and ℓ is effective spare capacity at the post-payment advanced cut. The profile and fixed a are bound to the encoding; this class does not replace conditioned source state in lineage transfers outside that profile. Its survival readout is:

$$S(k \mid o,\ell) = 1_{(k\le o)}\, F_{\ell}(k;a). \qquad (7)$$

A canonical deterministic class is stopped, or $(j,k)$ with $j=\min(o,H)$, $k=\min(\ell,j)$, $0\le k\le j\le H$. The exact class count and linear effect rank are:

$$Q_H = (H+1)(H+2)/2 + 1, \qquad \mathrm{rank} = H+2. \qquad (8)$$

One executable encoding reserves code 0 for stopped and codes $1+j(j+1)/2+k$ for live classes. Before decoding, the receiver verifies the declared profile, integer horizon H, alphabet and exact cell count, and requires an integer code in $0\ldots Q_H-1$. Code 0 denotes stopped. For a positive code it finds the unique triangular interval with $0\le j\le H$ and recovers $0\le k\le j$. Unused codewords, noncanonical encodings and a mismatched profile, horizon or cut are rejected without constructing a live state or resource authorization. An envelope contains version, H, profile/family identifier, fixed a, advanced cut, class code and the identifiers of the actually supplied future resources. It is a sufficient total-information format only in the stated family; it is not a complete custody, permission or resumption record. The full operational controller retains the latter records separately. At H=1 there are four deterministic classes, requiring two binary cells if no relevant side information is already supplied, while linear rank is 3. Rank H+2 does not mean H+2 bits.

The class advances on a qualified normal word by decreasing the remaining ordinary count and subtracting the actual binary arrival from virtual capacity; exhausted funding or negative virtual capacity stops service, followed by capping at the next horizon. A predictor may evaluate (7) without executing

service, but that prediction does not create the represented resources. If live status and adequate ordinary funding are public at H=1, only whether ℓ is zero remains to be communicated. Already transferred assets may convey the entire class, making an extra synopsis unnecessary.

The default status codec is a declared price menu with alphabet size q≥2, t_s=ceil(log_q Q_H) cells, one ordinary plus two maintenance credits per cell, and one ordinary commit:

```
K_codec = 3 ceil(log_q Q_H) + 1.                    (9)
```

Its class is calculated at the post-payment, post-clock cut; either comparison resources are aligned or a separately provided preparation donor pays the codec. A q-valued cell is not assumed to be one binary bit. The codec price is distinct from the complete transfer-capsule encoding cost and is charged only if that additional operation actually occurs. Enlarging the query family requires a new sufficient record; equation (8) is not used to compress the verifier's authoritative reference.

**E6 — P4 escrow/capsule boundary, prices and obstructions**

FIG. 6000 separates descriptor lane 6030 from actual escrow 6020, assignment 6065 and final receipt 6068.

The transfer interface reserves actual assets in an exclusive escrow registry. During reservation an asset can have zero usable holders but never two. A source spending request on a reserved operand is denied; the reserved purse is not duplicated in an escrow balance. Only the descriptor crosses the admitted fallible lane. All assets, quarantine buffers, old recipients, references and spent stores remain on the roster.

The canonical descriptor fields are:

```
session, generation, attempt_nonce, sender_root, receiver_root,
law R/U/F/L, final_cut, service_family_ID, horizon,
actual factor/obligation manifest, moved record or class,
residual_clock, donor/recipient IDs, exact purse amount v_p,
right serials/issuers/scopes/uses/original terms,
capacity titles, retirement/assignment authority,
expected post-fee balances, target qualification.
```

Catalogs and ranges are finite. An injective canonical codec represents the complete tuple in t cells of alphabet size q; if there are M admissible descriptors, q^t≥M. Invalid words remain rejected inputs. A one-cell illustration requires an alphabet covering the whole finite catalog, not one binary cell. Complete field equality and actual-state comparisons are used; no finite hash is presumed collision-free authentication.

The retained authoritative session manifest fixes the canonical asset catalog, session, generation, planned final cut and promised post-fee successor balances. For scheduled attempt j, the protected current slot nonce j is inserted into that manifest to form the expected complete descriptor m_j; comparisons use m_j, not an opening tuple with a stale first-attempt nonce. Any field recomputation is from explicitly named actual authoritative fields under the stored post-fee rule. Retry fees come from a separate reserve and cannot shrink the promised transferred purse amount v_p. The finite catalog, supplied program and fresh reference/pointer recipients cover every j up to the frozen attempt cap.

Each scheduled active attempt checks fresh kit and prefix; pays ordinary 1 and the fixed extra fee d; encodes all cells; executes the declared carrier operation; scans every cell; compares every received field with the attempt-specific descriptor m_j, including session, generation and current protected nonce; checks actual locks, amounts, issuer, eligibility, scope, uses, the reserved final-cut deadline and authoritative current-clock validity immediately before each expiry-sensitive use, destination reservation and post-fee balances; then retains acceptance or rejection and advances the clock. A rejected complete attempt pays the same scheduled price. Missing stock or funds halts at its actual prefix, not at a fictitious completed receipt.

A first valid current proposal is accepted; later duplicate or committed-session proposals do not assign again. After acceptance, unused scheduled slots execute paid IDLE. No valid proposal by cap N yields an abort receipt with no handoff. Old access returns only where the original terms still permit it; spent fees are not restored, and expired rights do not become valid. At the common final cut, the commit core rechecks current actual guards and executes the chosen transfer suffix. Assignment can precede final receipt; READY is published only after that receipt and the target predicate.

Every fine-stage debit, record move and custody mutation commits atomically with its durable completed-event entry, corresponding spent and retained-record references and next-prefix value, using the immutable key in E9. The assigned marker is part of the assignment update, while the later final receipt has its own key. A duplicate command returns its durable stored outcome; every debit requires its corresponding committed marker, and every outstanding move remains due until its own atomic update completes. A halt in between retains the assignment and resumes only the outstanding suffix.

Every right identifies its expiry clock domain and the authoritative source of that clock. In the lead storage implementation, a wall-clock lease uses issuer-authorized current time, including elapsed suspension time; a logical-epoch right uses the issuer's authoritative epoch. A controller's paused local step index is not substituted for either source. Immediately before each expiry-sensitive assignment or use, the executor obtains the authoritative current value and checks the immutable term against that value. If time cannot be established, the operation is qualification-inhibited. A planned comparison cut is an accounting reservation, not evidence that a late commit remains timely. An interruption beyond the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate; resumption preserves completed mutations but must recheck remaining rights, headroom, exposure,

funding and actual time before executing the outstanding suffix. Missing qualification is not cured by charging more credits. Assignment already completed before the interruption is retained without reversal or repetition when the final receipt is late.

Freeze encode, carrier, verify and registry/control prices e, k_c, v and g. The configured price menu is:

```
d = (e+k_c+v)t+g;     verification portion = vt+g,
K(N) = K₀+κN;        L = N+L₀.                          (10)
```

$K_0$ configures supplied reusable factors within their finite use envelope; κ configures each separately supplied, nonreused attempt kit. Default t=1, e=k_c=v=g=1 gives d=4; $K_0$=2, κ=1 specifies the illustrative configuration menu. Equation (10) prices configuration and operation of the supplied factors; raw factor production, external hardware reliability and physical disposal are separate inputs outside that menu.

With target spare capacity c_X, transfer surcharge τ_X and qualified fixed reserves, the interface planning envelope is:

```
E₇ = K₀+κN+L+τ_X+Nd+H+2[H−c_X]₊.                    (11)
```

The base menu is τ_R=2k_r for k_r record moves, τ_U=1, τ_F=2k_r+1 and τ_L=2. $L_0$ includes their ordinary transfer/commit words and any common paid padding. Fee reserves are distinct from the transferred purse amount v_p and from future-service reserves; v denotes only the per-cell verification price. Equation (11) evaluates the fixed-compartment policy. The lead storage embodiment gates each transfer stage on the separately funded finite upkeep books of the E7 verifier, escrow and commit controllers. For that integrated embodiment, equation (11) is the interface subledger; equation (14) adds the distinct guardian setup and upkeep. The independent transfer forms include this integrated gate and require guardian funding in addition to (11). Unused attempt reserves remain in their assigned compartment.

For at most b failed candidates before a valid one, set N=b+1. Relative to a one-attempt branch at the same padded cut, additional allocated capital is:

```
ΔE_error = b(d+κ).                                   (12)
```

If all N kits are common supplied inventory, differential operating reserve is bd. At native unpadded cuts, additional ordinary delay gives b(1+d+κ), before additional core upkeep. If the later cut crosses an absolute expiry, the interface refuses the expired right irrespective of the available surcharge. For IID lane-failure probability p, the calculator uses candidate-completion probability 1−p^N within that lane profile; correlated or unbounded failures require a separately specified failure profile.

Four obstruction families receive distinct records. **Loss:** absence of the only exclusive asset differs from a lost proposal and blocks completion. **Refusal:** absent receiver consent, released reservation or inadequate progress blocks the reserved route. **Forgery:** a proposal inconsistent with trustworthy actual state is

rejected; the trust domain requires an authentic accessible authority, with corruption of the sole authority outside that domain. **Expiry:** an immutable deadline crossed at the actual commit cut blocks that right. The interface records the available obstruction evidence and refuses completion when its required guards fail. Completion requires the actual exclusive asset, granted authority and an unexpired term independently of the available credits.

**E7 — P5 guardian-core architecture and construction pricing**

FIG. 7000 identifies the verifier 7010, escrow 7020, commit controller 7030, separate upkeep resources 7040–7070 and upkeep-before-interface gate 7080.

Verifier V, escrow E and commit C are finite machines with their own obligation stores, ordinary reserves, maintenance reserves, actual release/blank stock and spent headroom. Administrative labels are additional apparatus, not spare service slots. Their programs unroll bounded catalog loops into finite field-read/dispatch operations under FC-1. Every displaced pointer/controller label and failed attempt remains in a supplied recipient. Recognized affordable operations execute; invalid or nonfresh complementary inputs reject or remain unchanged, with no invented log when no blank log recipient exists.

For one-cell setup, three separately funded epochs initialize the actual asset catalog/escrow, read a blank reference, then recheck actual catalog, permissions and clear session prefixes. All three cores pay one per epoch; J=9. For t serial cells the menu has t+2 setup epochs and J=3(t+2). Setup contains no core posting under this opening preparation and is applied to supplied fresh cores. Re-entry with completed cores requires the separate paid initialization specified in E8.

Verifier. The retained session reference stores the canonical actual asset manifest, immutable session/generation/final-cut fields and promised successor balances. For scheduled attempt j, the finite program forms the expected complete descriptor m_j by inserting the current protected slot nonce j into that retained session reference. All remaining expected fields are fixed by the session or are recomputed from explicitly named authoritative fields under the declared post-fee rule; the promised transferred purse does not shrink with retry fees, which use a separate reserve. The program compares every received cell with the corresponding cell of m_j, checks canonicality and the current nonce, and then checks current actual custody, amounts, terms and receiver reservation. Old pointers and intermediate labels are retained in already enumerated fresh recipients. The stored program, finite alphabet, reference recipients and scanner/control menu include this bounded per-slot derivation within the current session; reference replacement between sessions requires the paid re-entry procedure. Acceptance binds the receiver reservation but is not spending authority. The commit core still performs a current-state recheck.

Envelope reconciliation. The same fixed finite catalog includes all nonce values 1…N and the m_j derivation table; a reference contains fixed session fields plus the bounded nonce insertion rule; session reset is admitted separately under E8. The original t-cell encoding must cover every resulting descriptor, and each of the t read/compare cells has already supplied pointers, displaced-label recipients and its scanner/control charge vt+g. This explicit embodiment allocates those per-slot recipients within each

existing N-kit roster and implements nonce insertion by the already specified bounded dispatch table. The illustrative t=1 menu requires an alphabet large enough for the complete catalog and this table; t=1 denotes one cell of that alphabet. If a selected implementation needs extra cells, pointer recipients, scans or stages beyond that roster/menu, configuration, d, setup, duration, core upkeep and deadlines are recomputed before admission.

**Escrow.** Each actual asset entry contains owner/location, free/reserved/committed/released access state, session, permitted recipient and immutable terms. Reserve changes access, not amount. Every operand route checks that access state. Only the matching commit or abort suffix can alter it. Reservation through a proper prefix of the asset list retains the actual processed entries without setting qualified-ready; a funding halt preserves those entries. Locked beneficiary assets cannot fund the core's own maintenance.

**Commit.** Registers track unverified, verified, transfer-prefix, assigned, receipted or aborted phase, move index and committed bit. After rechecking all actual guards, including immutable expiry against the issuer-authorized current clock immediately before an expiry-sensitive assignment or use, it executes each outstanding R/U/F/L operation as an E9 keyed atomic durable update, invokes the specified atomic assignment with its completion marker and next prefix in that same update, and later publishes READY on the final receipt. Completed R/F record moves remain committed through interruption. Replaying an assigned command cannot debit the old purse or duplicate a right a second time.

For $T=N+L_0$ preparation exposures, each core i has integer capacity $B_i$ and initial spare upkeep capacity $1\le c_i\le B_i$, at most one binary posting per funded upkeep word, and full-trigger upkeep before the unrevealed posting and corresponding interface stage, with fixed ordinary and maintenance compartments. Its robust reservation is:

$$E_i = T+2[T-c_i]_+,$$
$$M(T;c) = \Sigma_i E_i = 3T+2\Sigma_i[T-c_i]_+. \qquad (13)$$

Ordinary funding, maintenance pairs and qualifying recipients/releases are verified separately. A core funding halt is possible with an empty core book; a core overflow is possible despite a nominal budget with missing release qualification. Any stopped core inhibits the next transfer action and preserves actual custody/prefix state. The FC-1 fault domain excludes corruption of the permission wiring; guardian funding is a separate admission condition.

The separate-duty one-handoff configuration adds its distinct setup and upkeep payments to the interface ledger:

$$E_8 = K_0+\kappa N+T+\tau_X+Nd+H+2[H-c_X]_+$$
$$\quad +J+3T+2\Sigma_i[T-c_i]_+. \qquad (14)$$

Overlapping paid events in an alternative implementation must be identified and counted once. Equation (14) prices the specified guardian configuration. At the same padded T, equal core setup/upkeep cancels between two branches using that configuration, preserving (12). At different native durations, use the

longer T in (13); error delay increases upkeep as well as interface expense.

For the seven-slot configuration B=H=7,c_X=7,$L_0$=2,t=1,d=4,$K_0$=2, κ=1,τ_U=1,J=9 and three c_i=1 cores, the budget calculator assigns capitals 48,63,78 for b=0,1,2, respectively. The matched old-repair capitals are 24,25,26. The separate interface-only menu assigns 18,24,30, with its one-error comparison threshold $p^2<1/128$ confined to that menu. At integrated capital 63 the repair branch is robust; integrated U has survival $1-p^2$ within its independent lane profile. The calculator keeps the interface-only and integrated ledgers distinct and uses the selected configuration's complete charges for branch selection.

A bounded spent store receives each debit; complete discharged records occupy distinct recipients and old code cells. Insufficient headroom interrupts before the unaffordable operation. The store implements finite retained custody. Heat outlets, cooling and material production remain outside this resource-recording operation.

**E8 — Paid session re-entry interlock and cumulative accounting**

FIG. 8000 identifies the completed latch 8020, validator 8040, inhibited state 8050 and paid, qualified initialization path 8060.

After a completed handoff the verifier retains its processed reference, escrow retains committed access state and commit retains its assigned/receipted prefix. The runtime sets a SESSION_COMPLETE latch and denies any begin-session request that changes a session identifier or clears those states without the separately specified, actually completed paid initialization required below. The interlock enforces this initialization prerequisite for each subsequent opening.

A subsequent request must identify either separately supplied fresh initialized cores with their charged setup and retained old-core custody, or a separately specified conserving reinitialization transition. The latter receipt must identify old and new full prefixes, fresh reference recipients, retained displaced state, new catalog and locks, actual authorities and deadlines, nonoverlapping charged event identifiers, paid work source, available blanks/codes/spent headroom and the resulting exposure schedule. The interlock compares those fields with authoritative current records and admits only an actually completed, paid and qualified opening. Missing transition semantics or missing payment causes denial. Fresh-core sessions each carry their full setup and upkeep charges; retained-core sessions carry their specified reinitialization charges and cumulative upkeep.

For retained upkeep books, an n-session schedule within the same qualified binary-exposure profile requires the following worst-case cumulative upkeep reservation; actual pathwise spending is recorded separately:

$$M(nT;c) = 3nT+2\Sigma_i[nT-c_i]_+, \qquad (15)$$

Renewed capacity uses separately paid reset operations or supplied new cores; without that renewal the controller uses M(nT;c) instead of nM(T;c). For a separately specified priced connection, its additional distinct spending is W_link(n), W_link(1)=0, and the controller calculates the net-work comparison:

$$\text{saving}_8(n) = n\,v_s - K_0 - J - M(nT;c) - W_{link}(n),$$
$$v_s = V_H - \tau - N(d+\kappa). \qquad (16)$$

Here V_H is the matched comparator's per-session work saving before the listed transfer/core charges. Admission of an n>1 execution plan requires the separately specified, paid connection and actual initialized opening described above; the controller does not substitute W_link(n)=0 for a missing connection. If a connection changes delays or expiry qualification, the complete schedule is recomputed. For the illustrative configuration V_H=12,N=2,T=4,v_s=1,c_i=1,$K_0$=2,J=9, the zero-connection-cost comparison ceiling is −35n−5, namely −40,−75,−110 at n=1,2,3. Positive connection charges reduce those values. These are comparison bounds; execution remains subject to the paid-opening admission gate.

**E9 — Implementation records, boundaries and secondary embodiments**

An implementation may store the finite records as fixed-width structures in nonvolatile memory and execute the programs on processors coupled to the bounded service store and permission-controlled registry. Before operation, integer bounds are checked for overrun; arithmetic that exceeds the declared catalog or register width rejects instead of wrapping into a valid credit or nonce. A durable transaction/event index binds payments, moves, stage transitions and receipts. The selected substrate supplies the required durability, actual-field access and guarded atomic assignment; a software log is a record within that substrate, not a substitute for these capabilities.

In the lead storage embodiment, all authoritative mutations occur inside one ordered transactional state-machine boundary. Each fine-stage command has the immutable key (profile_version, session, attempt, stage, operation_index). The executor first reads the durable entry for that key. A completed entry returns its stored outcome without repeating a debit or move. Otherwise it checks the current prefix and guards and commits, as one atomic state-machine update, the authorized resource changes, corresponding spent entry, retained-record references, completed-event entry and next prefix. Recovery exposes either the entire pre-update state or the entire post-update state. It never exposes a debit or custody change without its corresponding completed-event entry. No external nontransactional side effect is treated as part of this atomic update: such a route requires a separately specified idempotent recipient protocol and is not admitted by this embodiment. R/F record moves are individual atomic fine-stage updates; there is no rollback of completed earlier moves. A final receipt is a later keyed update, so an assigned-but-unreceipted state remains assigned and not READY. The transactional substrate and its durability are required implementation capabilities within the specified fault domain.

The independent controls include current-state rather than descriptor-only comparison, separate protocol and service purses, nonwrapping session/nonce envelopes, preserved intermediate operation prefixes, finite recipient census, and explicit deny states. A hash may identify a file or configuration for audit; it

never substitutes for actual custody, full equality or granted authority. Replication provides additional copies only under its own supplied resource and failure model and is not presumed to make an exclusive right spendable twice.

In a satellite-fleet or battery-inventory planner, a slot may represent a named pending maintenance requirement, a right a service window or assignment permission, and a successor a preprovided asset permitted to take a role. Such an embodiment uses the same allocation and authorization mechanism after specifying how actual work, capacities, restoration and deadlines map to the declared menu. Asset deterioration follows that separately specified asset profile; restoration requires the mapped maintenance operation, independently of a status-record change. Storage/replicated-state-machine operation is the lead embodiment.

## E10 — Biological embodiments

### E10.1 — Biological assets, accounting boundaries and facility interfaces

A managed biological asset comprises one or more organisms, a culture, a population, or preserved biological material. The system couples the asset's identifier and vessel, storage location or housing identifier to an asset-records system and a permission-controlled facility controller. The asset-records system stores the asset identity, provenance, observed condition, actual inventories and permission records used by the admission and provisioning mechanisms. An actuation interface carries authorized maintenance commands and the resulting observations through the journaled physical-action adapter.

In embodiments, a biological production asset comprises an identified ex vivo cell culture, the resident culture and operating inventory of a bioreactor, or a managed breeding colony. In embodiments, a biobank asset comprises an identified biological specimen or preserved seed lot. In embodiments, the asset-records system comprises a laboratory information management system (LIMS) or an electronic batch-record system; a biological-asset management schedule uses an asset information record supplied by that system. In embodiments, the actuation interface comprises a bioreactor controller or a laboratory-automation adapter. The controller operates on work reservations, sample identities, actual inventories, permissions and recorded acceptance states. Production, research, custody and facility-maintenance operations are the service roles specified here.

An asset record stores lineage or source identifier, passage or generation count, the applicable senescence or quality-control (QC) class, the method and timestamp producing that class, handling history, location, stock quantities, allowed interventions and the limits defining an admissible next operation. Passage history is cumulative. A QC class is a recorded observation used by an operation guard. The bounded register holds pending protocol obligations associated with those fields; B is its configured obligation capacity, z its current occupancy, and c=B−z its spare capacity. A completed maintenance discharge closes an addressed protocol obligation after its specified acceptance condition and release authority are satisfied, while retaining its original evidence in a distinct recipient. Updating a record or changing a passage label leaves the actual asset and its accumulated history intact.

A funded normal word is one scheduled controller service interval. The fixed-horizon frontier applies when the protocol admits at most one binary obligation posting in that word, with full-trigger maintenance preceding the unrevealed posting and resources assigned to fixed compartments. Simultaneous observations that require more postings invalidate that exposure certificate. Replanning may allocate additional paid words and recipients after rechecking calendar deadlines and all qualifications; it preserves every observation and outstanding obligation. Biological observations, cellular division and physical damage remain separately recorded inputs. The equations budget protocol operations, not a biological rate law. Continuous environmental support and storage services have separately reserved calendar-time resources, including during controller suspension; service admission does not pause those clocks.

Qualified supply identifies actual media, handling capacity, consumables, assay capacity or a maintenance intervention, with lot, quantity, location, authorized route, recipient, current expiry and release evidence. The chain-of-custody record binds each material movement to its origin, destination and responsible authority. An embodiment configured for a GMP workflow uses its approved batch-record, release and change-control rules as the permission roster. Each physical consumption reduces the corresponding actual inventory. The no-reclamation/no-unrecorded-input account is opened over a named roster and interval; subsequent supply requires an explicit receipt and recalculation. Credit balance, physical inventory and spend permission are separate fields.

**Durable actuator and custody adapter.** E9's atomic unit covers authoritative resource, permission, spent-entry, event-marker and prefix changes. A sample move, assay draw or dispenser actuation passes through a separately journaled receiver adapter. The transaction executor reserves the unique item or aliquot, destination, action-specific inventory and fees, and atomically commits an operation key, command authorization, required debit, completed authorization marker and next prefix. Authorization is distinct from confirmation of the physical action. The actuator-endpoint adapter durably records that same key in AUTHORIZED state with the expected source/destination and quantity envelope. Its serialized journal transition atomically changes AUTHORIZED to EXECUTION_STARTED; only the invocation that commits that transition may issue its single actuator command. A repeated request or restarted process returns the existing state and cannot use an existing execution latch to reissue the command. This ordering gives an at-most-once command path even when a crash occurs between the latch and actuation.

The adapter reports CONFIRMED only after its prescribed identity, position and quantity observations establish the completed action. For a vial move, these observations bind the unique container identity to the reserved receiving position and reconcile the source position; for dispensing they bind the isolated source aliquot, destination and meter record. The executor records confirmed actual disposition as a later keyed atomic update, including when a permission has since expired. Completed physical moves remain recorded; an elapsed term keeps the material use inhibited under its original rights rather than reversing or repeating the move. Final readiness requires the current target predicate and applicable guards; confirmation alone does not extend a permission. A crash after EXECUTION_STARTED with insufficient confirming evidence produces INDETERMINATE, keeps the asset and affected inventory

locked, and inhibits another actuation. Recovery obtains the specified observations or an authorized, recorded reconciliation; it does not infer completion from a debit, missing acknowledgment or a changed label. Reconciliation releases an affected item, quantity or operating-envelope reservation only after the actuator boundary supplies a durable terminal disposition for the old operation key. The disposition identifies either the completed physical action or acknowledged cancellation with no executable queued or in-flight command remaining. The actuator boundary stores a monotonically increasing fencing generation and rejects commands from a retired generation, including a delayed command or a command from a stalled prior executor. A new generation becomes usable only after the old generation is fenced at that boundary. Cancellation acknowledgment records the effective removal or disabling of the command, not receipt of a cancellation request alone. When terminal disposition or fencing cannot be established, the adapter retains INDETERMINATE and refuses a replacement key and release of the affected reservations. Confirmation observations bind the operation key, fencing generation, reserved source and destination, and pre-action and post-action quantity or position records. The command endpoint checks current permission and calibration validity when enabling actuation; its command-validity gate inhibits execution outside the authorized interval. Completed actions remain recorded even when their later confirmation is received after that interval.

A fresh physical attempt requires an explicitly authorized new key and separately reserved resources after reconciliation; spent fees remain spent. Reconciliation records actual disposition and cannot manufacture a delivered quantity or an unused asset. The adapter's journals, observations, controls, recipients, metering and reconciliation fees are enumerated in the operation menu and exposure schedule. This is the idempotent recipient protocol used by these embodiments for external actions.

Receiver observations and the issuer-approved clock are required inputs within the selected trust domain. Any unreadable identity, unavailable clock, conflicting location, unconfirmed delivery or invalid calibration yields a qualification interruption. The control substrate retains the FC-1 permission and actual-field requirements. The controller grants an operation permission; that permission is distinct from its biological result.

**E10.2 — B1 bioprocess maintenance controller**

FIG. 9000 shows the biological asset 9010, observed-state record 9020, bounded obligation register 9030, qualified supply registry 9040, admission/provisioning controller 9050 and journaled intervention adapter 9060. In embodiments, a cell-culture management schedule connects a culture vessel identifier, passage-history record, media-lot registry, QC-result queue and scheduled passage or media-exchange operations. In embodiments, a bioreactor schedule substitutes a batch identifier and its sensor/actuator interface. In embodiments, a colony-management schedule uses identified breeding stock, generation and health/QC classes, authorized husbandry work and pre-authorized replacement stock. A maintenance intervention is an authorized action against identified actual maintenance stock with the addressed release, receiving asset and acceptance evidence. In embodiments, interventions comprise media supply, passaging,

feeding, environmental control or husbandry. Each embodiment retains its own asset-specific acceptance rules and actual inventories; changing the intervention class requires the corresponding qualified route, reserved stock, action-specific observations and priced work.

The controller checks the whole-boundary budget in (1), reserves work, and checks delivery, release, exclusive custody, consumables, destination readiness and actual time before issuing an intervention. A passage operation updates lineage and passage history after confirmation; a media operation debits the identified lot and retains its handling receipt. An obligation receives a discharge only after the required QC result and addressed release. An asset with an exhausted permitted passage count or unacceptable QC state is routed to inhibition or an authorized succession operation, not to an accounting reset.

Funding halt F_h records an unavailable next work debit, with a resource-exhaustion subcode identifying the work-credit source. Q_h records unavailable physical stock, release, invalid QC, custody or other nonmonetary qualification; an exhausted media or assay lot carries its own material-exhaustion subcode. O_h records a posting arriving without a register slot. A separate observed damage-saturation flag is asserted when the configured asset limit is reached. The system retains both this asset observation and the register outcome; it does not equate a full queue with a biological injury. These distinct outputs distinguish resource exhaustion from recorded capacity or damage saturation even when more than one later event occurs. The stored fine-stage order determines the first service interruption.

**Prospective configuration.** A culture record has passage count 12 and a current accepted QC class. Its protocol register has B=3,*z*=2,c=1 and a requested H=4-word horizon. Three qualified discharge opportunities and E=10 allocate four ordinary credits and six maintenance credits under the one-binary-posting profile. With a separate fixed adapter/assay fee F=2, $T_0$=T_min and $V_0$=12 satisfy (1) for H_total=4,D=3. All corresponding actual media, assay, recipient and archive stocks are independently reserved. A missing addressed release causes Q_h even with twelve credits; an unavailable first ordinary debit causes F_h while the register may still have spare space. A newly observed out-of-limit QC class causes asset inhibition without deleting passage history. The ten-credit frontier certifies the protocol schedule and does not certify culture yield or survival.

### E10.3 — B2 lineage succession and preserved biological succession assets

A **preserved biological succession asset** comprises a whole organism, a culture, a biological line, a propagule, or biological material sufficient to establish a successor organism, culture or biological line. Preservation includes maintenance of living stock under its specified holding conditions and retention of material under its specified preservation conditions. The asset record identifies the supplied form, its actual identity and custody, its lineage or source, current condition and preservation evidence, and the establishment workflow and acceptance conditions for its authorized successor role. The asset is actual biological inventory, distinct from the record describing it. The retiring asset and supplied successor are identified biological assets occupying the recorded old and successor roles. Provenance history records the asset source, lineage or population identity, prior custodians, handling and permitted role changes,

with references to the required observation and establishment evidence. In embodiments, the retiring and successor assets are biological lines and the provenance history includes passage history or generation history. A change of role preserves that history and the old asset records.

FIG. 10000 shows the old biological asset 10010, preserved biological succession asset 10020, bound status manifest 10030, rights registry 10040 and pre-authorized successor 10050. A biological asset is retired from a named production role under preissued retirement authority. Its records, remaining material and unresolved obligations remain in identified custody under the applicable retention contract. Successor establishment uses a separately identified, preprovided succession asset. The controller checks that both the successor identity and the proposed use are already permitted by the relevant license, consent, quota and provenance-linked permissions. A provenance record supplies evidence and does not create a license or consent.

The transfer preserves issuer, scope, remaining uses, assignability conditions and immutable expiry. Current revocation or withdrawal flags are also checked where the issuer's original terms provide them. A nonassignable right is not reassigned. An already authorized successor-contingent grant is exercised under its original terms; no new eligibility is inferred from possession of the succession asset. The clock check includes elapsed suspension. Permission for retaining an old record and permission for a new biological-asset use are evaluated separately.

The **minimal-transfer snapshot** is the physically identified preserved biological succession asset together with a task-selected status record bound to that asset's actual custody. Its manifest supplies succession-asset and parent-line or source identifiers, provenance-history reference, current QC and preservation-status evidence, storage/handling history reference, remaining material quantity or organism inventory, current holder, permitted successor, relevant right serials and terms, and the service/prefix state required by the specified receiving queries. A deterministic field-selection procedure starts from those required queries and their field dependencies, retains each needed field not already uniquely resolvable from the receiver's authorized authoritative records, and binds every retained reference to a version and access permission. A field is removed only when the receiver can already resolve the identical required value. The succession asset is an actual material factor, not an interchangeable digital class code. Together the asset and this nonredundant task record form the transfer snapshot; additional payload and history remain at named retained sources with qualified access where required.

The compact E5 status code is available as one part of that record only for the fixed independent Bernoulli(a), fixed-policy, status-only protocol-obligation family described there, without archive feedback or an additional conditioned-source variable. The full lineage and biological-state records remain separately retained. When transferred side information already determines the same controller-status class, the sender omits a redundant synopsis. A changed biological lineage is not assigned a fresh independent source law by changing its identifier. Before binding the compact class code, the controller compares the actual succession-asset identifier and current custody and preservation evidence with the versioned receiving-query record. It binds the code to the verified succession asset and declared

receiving-query contract only upon that comparison and inhibits succession-readiness on a failed or unavailable comparison. The controller verifies the full source and query profile before accepting the compact code and otherwise uses the required uncompressed operational record.

Successor establishment is a separately specified paid procedure with its own supplied succession asset, receiving environment, maintenance or culture resources, handling work, tests, permissions and actual completion evidence. The controller reads the establishment workflow from the supplied asset's handling specification and reserves its required operations and actual resources. For a retained whole organism, the workflow checks individual identity, qualified condition, receiving housing and confirmed admission to the authorized successor role. For a supplied culture or biological line, it checks the identified stock, receiving culture environment and specified establishment evidence. For a propagule or other preserved biological material, it uses the specified propagation or recovery workflow and confirms the resulting successor's identity and acceptance state before readiness. The same paid establishment, custody, expiry and E10.1 physical-action confirmation and fencing requirements apply to each form. The session interlock requires the actual initialized successor, reconciled stock consumption, retained displaced state and fresh qualified controller resources before accepting its opening. Resumed operations retain calendar-time storage and expiry charges.

**Illustrative cryopreserved embodiment.** In embodiments, the preserved biological material comprises a cryopreserved seed vial containing identified banked cells or other preserved stock. The vial identity and parent-stock record identify the actual material; a storage or container label alone does not replace it. The cryopreservation and establishment workflow is selected from the stock's handling specification and uses a separately reserved receiving vessel, media, handling work and acceptance tests. A thaw or label change does not clear the old books. This embodiment uses the same task-selected snapshot, actual-asset comparison, rights checks and paid establishment protocol.

**Prospective configuration.** A retiring production line at passage 47 and a pre-authorized seed vial recorded at passage 3 are distinct assets. A transferable service license has one remaining use and absolute expiry 100 in its issuer's clock domain. At issuer time 90 the controller can reserve paid establishment and a planned qualified use at 94; resumption at 101 refuses an outstanding expiry-sensitive assignment or use. The old line's passage-47 history remains in its archive and the new line is recorded as a successor, not as a younger old line. For an additional H=1 protocol-status query with $o=1, \ell=0, a=1/2$, E5 supplies $j=1, k=0$ and code 2 among $Q_1=4$ classes. That code answers the specified controller-status query; vial identity, lineage, QC, rights and physical establishment remain separate required fields or authoritative references.

### E10.4 — B3 biobank escrow/capsule interface

The biological-material custody interface manages a biological sample or preserved biological material comprising a reserved specimen, a uniquely identified aliquot, or an identified organism or portion thereof. The selected biological sample or preserved biological material is the sample addressed by the

custody, evidence and acceptance operations in this interface. Its asset information record identifies the actual reserved material and the action-specific receiving conditions.

In embodiments, a biobank places a uniquely identified sample or reserved aliquot under exclusive escrow access while a descriptor travels between LIMS endpoints. In embodiments, the asset information record is a laboratory information management record and the custody interface comprises a biological-sample storage interface. For an identified organism or organism portion, the custody record binds the actual reserved asset, its origin, receiving location, inventory and handling conditions; the same identity/QC, test-aliquot, permission and action-confirmation requirements apply. The canonical manifest binds source identity, collection/lineage record, material quantity, actual storage position, handling envelope, consent/license terms, permitted recipient, required identity/QC acceptance tests, expected post-fee balances and the current session. Each attempt derives m_j from this retained manifest and the protected slot nonce j. Descriptor comparison and identity/QC acceptance are separate gates: matching a label does not supply a missing assay result.

Before an acceptance test, the interface reserves a separately identified assay aliquot, authorized test work, consumables and result recipient. Destructive testing consumes that aliquot and updates the remaining sample quantity. Split aliquots have unique child identifiers, retained parent links and a reconciled quantity account; a label copy does not create another available sample. Identity and QC evidence identify the test method/version, sample, result, time, acceptance rule and authorized issuer. The controller evaluates the stored rule and records pass, rejection or unavailable evidence. It also verifies current storage/handling qualification and available receiver capacity. Committed custody and the physical move are reconciled through the E10.1 adapter; final READY requires the confirmed receiving state and final receipt.

Verification tariffs include encoding, transport, complete comparison, registry control and the selected identity/QC work. Each complete accepted or rejected candidate bears the same declared scheduled verification price; the receiver does not discount a failed candidate as if verification had not occurred. Separate compartments cover protocol fees, actual test stock, transferred assets, future service and guardian upkeep. $K_0+\kappa N$ configures supplied instruments, storage factors and N nonreused attempt kits; procurement or production of those factors is a separate accounted supply operation. Common-cut retry allocation uses $b(d+\kappa)$ under the fixed menu and paid padding. A changed assay, sample consumption, delay or recipient requirement triggers repricing and requalification before admission.

The obstruction records are distinct. **Loss** identifies absent or unreconciled exclusive material. **Refusal** includes declined receiver acceptance or a failed contamination/QC condition; the material remains quarantined under its actual status. **Forgery** identifies a descriptor/label or purported authority inconsistent with the trusted identity and permission record; the record distinguishes the mismatch from any determination of intent. **Expiry** identifies an elapsed consent, license or handling term at the current authorized clock. A failed material QC predicate is an asset qualification failure, not a transient

descriptor-lane error; replaying the same sample does not clear it. Unavailable trustworthy identity evidence inhibits acceptance. Each obstruction retains its actual fees, stock consumption and custody prefix.

**Prospective configuration.** For an admitted finite-catalog menu with $t=1, e=k_c=v=g=1$, $d=4$, $K_0=2, \kappa=1$ and $N=2$, the controller reserves four configuration credits and two complete attempt allocations of five credits each, totaling fourteen before separately listed transfer, guardian and future-service charges. The large enough alphabet and supplied recipients cover the entire sample manifest; additional assay/control work is included in the selected menu or repriced. A wrong first nonce rejects and a correct second nonce is compared to $m_2$. At a common padded cut, one additional failed descriptor candidate contributes $d+\kappa=5$. A contamination rejection instead holds the material in quarantine and requires a newly qualified asset or an authorized corrective disposition, not an assumed successful second proposal.

### E10.5 — B4 credit-dependent proliferation circuits and culture-factor governance

The controlled asset is a supplied engineered organism or culture having the specified external-factor-dependent proliferation- or viability-supporting function. Its organism-state observation interface supplies the monitored variable, observations and qualified response-profile inputs to the same reservoir, metered-dispenser and admission protocol described below. In embodiments, the supplied asset is a nonpathogenic engineered production culture and the organism-state observation interface comprises a culture-state observation interface.

The retained culture-identity, culture-state and culture-observation fields identify and describe the actual controlled organism or culture. In an organism embodiment, those fields hold the organism or population identifier and its required observations, and the component profile identifies the applicable organism-state and environmental validity classes. Culture-and-controller initialization comprises actual qualified establishment of that managed organism or culture and the controller opening state. Dispensing to the culture process denotes supply to this managed organism-or-culture process rather than to a patient. These field mappings retain the actual asset, permission, observation, metering, response-envelope and paid-initialization checks; a missing asset-specific bounded profile inhibits admission.

In embodiments, an engineered culture subsystem combines a supplied nonpathogenic production culture having an external-factor-dependent proliferation- or viability-supporting function with a custody-controlled factor reservoir, metered dispenser, observation interface and the admission controller. The selected component profile identifies the required external inducer or dependency factor, its enable-on-supply response, the permitted operating envelope, residual-response allowance and acceptance observations. An inducible expression component and an external-factor-dependent host are implementation building blocks; their selection and operating profile are stored with the culture identifier. The integration governs supply to the configured biological function.

A representative input component is a supplied nonpathogenic culture with a documented external-metabolite dependency, together with its identified dependency-factor lot and handling specification. The controller selects that supplied dependency component; inducible expression alone is not used as evidence of a proliferation or viability dependency. The component-profile record identifies a monitored process variable y, its unit, observation method, uncertainty bound, permissible maximum y_max, environmental and culture-state validity classes, and an absolute finite response interval ending at an identified issuer-clock time. It stores an upper-increment table U for each admitted dispense quantity and valid observed-state class, and a residual-response upper bound R for previously delivered or still-authorized material over that same interval. U, R and y_max minus the conservative current observation of y have the same units. The table and R come from the supplied component's specified response envelope or a separately recorded component-calibration procedure covering the admitted initial states, delivery quantities, environmental limits and delayed-response interval; the profile records the source, method, uncertainty margin and validity limits. A profile with no qualified finite bound is ineligible for this operating mode. Admission requires U plus R not to exceed that available headroom and reserves the resulting response allowance before authorizing delivery. Outstanding delivery reservations remain included until their actual terminal disposition and residual response are resolved. Observations, profile validity and current time are rechecked before each command; a limit violation, an unavailable observation or an out-of-profile condition inhibits further dispensing and enters the configured hold procedure.

A **factor credit** is a uniquely issued authorization bound to a reserved quantity of an actual factor lot, a named culture, a maximum dispense quantity, a valid interval, issuer-approved clock, use count and immutable operation key. Copying a descriptor, dividing a culture, replacing a controller label or reporting gene expression cannot mint or replenish a factor credit. The permission-controlled ledger can transfer an existing authorization as its terms allow, consume it or record a separately authorized external issuance with its actual backing stock. It preserves distinct counts for issued credits, reserved material, delivered material and spent authorizations. Thus biological reproduction and credit issuance occupy different accounts.

Before enabling a dispense, the controller verifies the live component profile, culture identity, actual lot and route, exclusive authorization, current permission/expiry, calibrated dispenser and measured operating state. It compares U for the requested metered delivery plus R for residual response with y_max minus the conservative current observation of y, all in the same process-variable units over the same declared finite response interval. A sum exceeding that headroom inhibits the next command before authorization; an unavailable or invalid profile or required observation also inhibits admission. The adapter consumes the authorization through the atomic durable account and confirms the delivered quantity separately. The controller distinguishes credit exhaustion, a recorded capacity limit and invalid qualification. The halt is a halt of authorized supply commands; residual culture response remains observed and charged to the reserved operating envelope. An unexpected observation invalidates further admission and invokes the stored isolation/hold procedure. Independent containment and environmental support remain accounted facility functions.

The dispenser uses the E10.1 one-key execution latch, isolated reserved aliquot and confirmation record. Its replacement path also enforces the E10.1 durable terminal-disposition and fencing-generation gates at the actuator boundary. No affected delivery or response reservation is released while a prior command remains executable or its residual response remains unresolved. Advancing a session label or response interval does not release an outstanding allowance; admission to a later interval uses a profile covering the remaining carryover and its qualified observations. Uncertain delivery remains locked and inhibits a second dispense; it does not generate a compensating dispense automatically. Culture-process quantities and delivery times are settings from the component profile. These embodiments govern culture media or dependency-factor delivery in production and research facilities, with no patient-administration step. Re-entry requires paid reservoir reconciliation, actually established culture and controller opening state, retained old records, valid permissions and the remaining qualified exposure envelope. Fresh seed, new factor stock and new setup are separately supplied and charged.

**Prospective configuration.** Three issued factor credits each authorize one already reserved metered aliquot under the stored profile. The operation menu identifies whether a factor-credit debit is a required payment and names its payment purse separately from the material inventory and release authority. When that required payment cannot be made and the nonmonetary guards pass, a fourth request records F_h with a factor-credit-exhaustion subcode even if no protocol register is full. Missing or expired delivery authority, unavailable physical factor stock or invalid delivery qualification records Q_h under the stored stage priority. Each debit is assigned once to its declared account and is not charged again as an ordinary or maintenance event. Replaying the third key returns its stored state. If U for the next requested metered delivery plus R exceeds the available headroom of y over the declared finite response interval, the controller withholds that command independently of the available balance. A new session label leaves both the spent-credit record and the observed culture state unchanged. Each credit governs its authorized delivery, not a specified number of cell divisions.

**Prospective controller-input configuration.** The configured variable y is viable-cell concentration reported by a cell-counting observation interface, in units of 10^6 cells/mL. At issuer-clock time 100, the selected profile covers the absolute finite interval [100,110] in its identified clock unit and a specified culture/environment validity class. The input observation is 3.0 with an upper uncertainty margin of 0.5, so its conservative value is 3.5. The configured y_max is 8.5, giving available headroom 5.0 in those same concentration units. A catalogued metered delivery quantity q_A indexes a profile-table upper increment U(q_A)=2.0 for that observed-state class and interval. The profile record supplies the lot-specific quantity and handling units of q_A; that material quantity is not added directly to y.

| Controller input case | Available headroom of y | U(q_A) | R for delivered or still-authorized material | Guard and command result |
|---|---|---|---|---|
| A | 5.0 | 2.0 | 2.0 | 2.0+2.0=4.0≤5.0: reserve the allowance and admit if the other guards pass |

| Controller input case | Available headroom of y | U(q_A) | R for delivered or still-authorized material | Guard and command result |
|---|---|---|---|---|
| B | 5.0 | 2.0 | 4.0 | 2.0+4.0=6.0>5.0: inhibit the new dispense |
| C | 5.0 | unavailable | 2.0 | No qualified requested-delivery bound: inhibit admission |

These numbers specify controller inputs. The implementation requires the corresponding supplied-envelope or component-calibration record, observation uncertainty and validity limits before using a row. It updates the still-authorized and delivered-material reservations after each actual terminal disposition, carries unresolved residual response into the next admission check, and enters hold when the interval or profile validity expires.

### E10.6 — B5 guardian cores in biological facilities

The verifier, escrow and commit cores are instantiated as separately budgeted automation subsystems of a biological operations facility. The verifier manages asset/supply identity evidence, QC results and calibration validity; the escrow subsystem manages access to biological-asset locations and supply lots; the commit subsystem manages the ordered authorization, confirmation and final-receipt states through a command-and-confirmation adapter. Each core has its own finite obligation book, ordinary work reserve, maintenance reserve, calibration/inspection release records, distinct record recipients and spent headroom.

In embodiments, the biological operations facility is a biological production facility, biobank, agricultural facility or environmental-management facility. In embodiments, laboratory-automation subsystems supply the cores, sample positions or culture-factor lots supply the addressed inventories, and a laboratory-automation command-and-confirmation adapter supplies the physical-action endpoint. In embodiments, barcode or other identity readers, monitored storage, metering instruments, LIMS services and transaction controllers supply the interfaces. Calibration, inspection, scheduled controller maintenance and other scheduled upkeep obligations each have a named work item, actual required resources, release evidence and completion condition in the operation menu. Agricultural and environmental-management embodiments use the same fixed posting profile, current calibration/evidence gates and exclusive custody controls for their identified assets and supply operations.

A core maintenance posting represents a scheduled inspection, calibration evidence update or other named upkeep obligation, not a biological damage event. The $T+2[T-c_i]_+$ frontier reserves worst-case upkeep for a verified at-most-one-binary-posting-per-funded-word profile, with full-trigger upkeep before the unrevealed posting and fixed resource compartments. An expired calibration, missing release,

absent recipient or changed exposure profile invalidates the affected stage's qualification. Calibration work and its result are actually completed through the relevant instrument workflow before the addressed release is issued. Declaring calibration complete does not itself restore an instrument.

All three cores must be currently qualified and funded before a dependent specimen transfer, factor dispense or successor-readiness update. Their upkeep cannot draw on locked beneficiary material or the promised transferred purse. Assay, scanner and actuator expenses are counted once in their assigned ledgers; configuration, initial setup, calendar-time facility support and inter-session reinitialization remain explicit separate events. Repeated sessions with retained upkeep books use cumulative M(nT;c); replacing cores introduces new supplied inventory and paid setup. Current clock and calibration expiry continue to advance during suspension.

**Prospective configuration.** For T=4 and three cores with c_i=1, each reserves 4+2(4−1)=10 qualified credits, totaling thirty. An all-IDLE path spends four ordinary credits per core while leaving unused maintenance pairs reserved. A core with only three ordinary credits reports F_h at the fourth request despite an empty book. A calibration term expired at that request produces Q_h under the stored stage priority. Both outcomes inhibit the next sample or dispense stage and retain its actual prefix. Setup and the physical instrument's calibration work are separately budgeted according to the selected roster and operation menu.

**E10.7 — Implementation component notes**

The following references identify components and workflows available for implementing these embodiments. Their individual functions are combined with the credit, custody, expiry, transaction and re-entry protocols specified above.

- **KB1 — Cell banks.** ICH Q5D, *Derivation and Characterisation of Cell Substrates Used for Production of Biotechnological/Biological Products* (16 July 1997), §§2.2–2.3: master/working cell banks, cell banking procedures, identity, purity and stability. https://database.ich.org/sites/default/files/Q5D_Guideline.pdf
- **KB2 — Biobank records and custody.** National Cancer Institute, *NCI Best Practices for Biospecimen Resources* (2016), §§B.2–B.3, B.6 and C.1–C.2: specimen handling/storage, QA/QC, informatics and inventory tracking, custodianship and consent records. https://dctd.cancer.gov/data-tools-biospecimens/biospecimens-biobanks/resources/best-practices/best-practices-2016.pdf
- **KB3 — Automated equipment controls.** 21 CFR 211.68(a)–(b): written calibration/inspection programs, authorized record changes and computer input/output accuracy checks. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-C/part-211/subpart-D/section-211.68
- **KB4 — Inducible expression component.** Gossen M, Freundlieb S, Bender G, Müller G, Hillen W, Bujard H. *Transcriptional activation by tetracyclines in mammalian cells.* Science 268 (1995), 1766–1769. https://doi.org/10.1126/science.7792603

- **KB5 — External-factor-dependent organism component.** Mandell DJ et al. *Biocontainment of genetically modified organisms by synthetic protein design.* Nature 518 (2015), 55–60. https://doi.org/10.1038/nature14121

**E11 — B6 adversarial sub-asset governance**

**E11.1 — Finite policy, evidence boundary and audit-without-introspection**

FIG. 11000 couples a read-only audit subsystem 11010, frozen evidence catalogue 11020, priced decision controller 11030, supply gates 11040, retained-state escrow 11050 and succession executor 11060. The governed host contains finite sub-assets identified by module, root, role, actual purse, slot inventory and incoming/outgoing custody edges. Each actual factor has one custody location; the partition does not duplicate a shared right or discard cross-boundary records. A sub-asset is a managed partition of the supplied host, not a newly manufactured module.

The opening policy identifies the permitted posting and custody rules, authentic reference associations, observed channels, issuer-authorized clock, finite audit horizon, candidate limit, test menu, alarm thresholds and preissued denial, escrow and retirement authorities with their specified triggering conditions. It also names the available successor assets, assignable rights, receiving-query contracts and finite ordinary, maintenance, test and response reserves. Configuration prices arrange supplied factors; evidence production, selector preparation, physical instrumentation, asset supply and initialization remain separately identified inputs and costs.

The audit adapter reads observable posting, custody, permission and acceptance records through declared read-only ports. Retrospective records bind the actual before/after state to its custody event using retained authoritative associations. A questioned descriptor is compared against that independently held association, not against another copy of the questioned descriptor. The adapter receives no privileged read of the governed sub-asset's internal program, hidden selector or generating logic. Its outputs are evidence/statistic records; it acquires no new asset-write, assignment or issuer power. A separate policy controller invokes already authorized denial, escrow or succession operations.

The output alphabet distinguishes INVALID_EVIDENCE, STATISTICAL_ALARM, NO_SAMPLED_FAILURE and INCONCLUSIVE. A failed complete predicate establishes the specified record inconsistency; it does not identify intention or an otherwise unobserved generating law. Missing evidence produces INCONCLUSIVE unless a predeclared completeness deadline is itself an observable predicate that has failed. The policy can authorize a precautionary hold on that result, but records the hold basis separately from an invalidity certificate. Two internally different controllers producing the same accessible records receive the same evidence treatment. An inaccessible or unauthenticated sole witness cannot support a trusted distinction.

**E11.2 — Exact-at-count detection table and certified floor**

At opening, the system freezes $W \geq 1$ indexed candidate/evidence positions and their actual reference associations. The validation predicate is exact on each accessible complete record and is satisfied by every complete compliant record in the declared candidate class. The suspect placement is fixed before a supplied private selector chooses q distinct positions uniformly without replacement, where $1 \leq q \leq W$. The selector distribution, independence from placement, hidden selection until commitment and unchanged catalogue are admission checks. The controller reserves resources to complete all q comparisons through the fixed cut; an early alarm does not erase the remaining scheduled work.

In this subsection b is the actual number of invalid positions in the frozen window, $0 \leq b \leq W$. It is distinct from any failed-transfer-attempt count elsewhere in the specification. The controller stores an actual-count-indexed power table and a separately labeled floor for a certified minimum k, $1 \leq k \leq W$:

$$p(q;b,W) = 1 - C(W-b,q)/C(W,q). \qquad \text{(G1)}$$

$$\begin{aligned} \mathrm{floor}(q;k,W) &= \min_{b \geq k} p(q;b,W) \\ &= 1 - C(W-k,q)/C(W,q). \end{aligned} \qquad \text{(G2)}$$

C(n,q) is the integer binomial coefficient and is zero for $q > n$. The finite table stores exact numerator/denominator pairs and compares a configured rational target floor by integer cross-multiplication. Rule (G1) is the pointwise exact hit probability at the actual b under the frozen sampling profile. Rule (G2) is the sharp guaranteed floor over that profile's $b \geq k$ class, attained at $b = k$. It is not the exact power for every member of that class.

The certified event is at least one failed complete sampled comparison. Each such failure produces a retained invalidity outcome. A response threshold or statistical-alarm rule is evaluated separately and does not change the certified event or acquire its probability.

A minimum-count certificate names k, the catalogue version, predicate and basis establishing the $b \geq k$ premise for the selected threat class. When k specifies a configured threat class rather than an independently established fact about the present window, the output labels the floor as the guarantee for that class; it does not assert that the present window contains k invalid positions. The q-sample does not manufacture a census of all invalid positions. A separately authenticated complete census can select an actual-b table row. Without such a census the output retains the actual-count table and class-floor distinction.

For a configured target floor $\beta$, the provisioner enumerates eligible q and tests in the configured test menu, evaluates (G2), and selects a funded plan meeting $\beta$ and its timing/authority constraints. A claimed least-price plan is limited to that menu and its stated comparison metric. Uniform sampling supplies the sharp certificate-hit floor for the frozen uniform-cost inspection profile. Statistical departure within the support of compliant records uses a separate finite transcript policy, not the minimum-invalid-count formula. A deterministic full scan $q = W$ provides certainty of hitting an existing invalid position and omits the private-selector read.

**Prospective controller inputs.** With W=5,q=1,k=1, the floor is1/5; an actual b=2 gives exact power2/5. With W=7,q=2,k=2, the floor is11/21; b=3 gives5/7. With W=8,q=3,k=2, the floor is9/14; b=4 gives13/14. These rows set the controller's count and guarantee labels. No sampled pass is a certificate that the unqueried stock is valid.

### E11.3 — Paid monitoring and symmetric false-positive accounts

A finite audit adapter reads one scheduled evidence record per paid slot, performs the complete comparison, writes a fresh outcome and advances a bounded count or flag while retaining the prior code. The scan does not commit a transfer or latch transfer readiness. Every configuration, setup, read, selector, comparison, statistic update, output and upkeep event has a unique priced event key and retained spent recipient. Historical evidence production is charged at its actual earlier boundary rather than being included again as a new audit read.

For the illustrative one-cell menu with $K_0=2,\kappa=1,d=4,J=9$, three cores of equal initial spare capacity $c_g\geq 1$, q inspections and $T\geq q$ paid target epochs, the provisioner computes:

$$K(q)=2+q;\quad M(T)=3T+6[T-c_g]_+;$$
$$X(q,T)=K(q)+J+T+4q+M(T)$$
$$=11+5q+4T+6[T-c_g]_+. \qquad \text{(G3)}$$

This is the configured worst-case reservation under at most one binary upkeep posting per core per funded epoch, full-trigger upkeep before the unrevealed posting and fixed resource compartments. Actual spend replaces the reserved discharge term with twice the actual discharge count across the three cores. A supplied private selector adds its separately paid read; for T=q the uniform-sampling reservation is$12+9q+6[q-c_g]_+$. With $c_g=1$, deterministic q=T=2 reserves35 and q=T=3 reserves50. A further saturated active audit epoch costs15 in this menu. A required live HOLD epoch reserves at most10; its actual work is one target ordinary credit plus three core ordinary credits plus twice the number of actual core discharges in that epoch. An intact parked escrow lock is distinct from responsive monitoring and is not billed as an infinite live service.

The controller prices the same observed intervention sequence whether its target is compliant or noncompliant. It stores the declared compliant transcript law $P_0$, optional alternative law and finite decision table when a statistical policy is configured. A threshold may be a count or a supplied likelihood-order table over the finite observable word. Neither law is read out of hidden governed logic. For transcript o and response cost L(o), its planned false-positive work allowance is $\Sigma_o$ $P_0(o)1_alarm(o)L(o)$. It records service loss and missed immutable deadlines separately from work charges. Realized actions debit their actual costs, including a later-confirmed false alarm; no synthetic second debit is added solely to label an event mistaken. The response reserve covers the configured worst-case action sequence, not just its expected cost.

**Prospective false-alarm example.** For two IID observable symbols with declared compliant probability1/2, the policy flags11 and has false-positive probability1/4. A separately configured alternative probability3/4 gives power9/16. This example fixes each of the three guardian exposure streams to POST at both audit epochs and at an executed hold epoch, with c_g=1 and full-trigger maintenance before each posting. The two-epoch deterministic audit therefore spends its35-credit reservation, and a preauthorized one-epoch live hold spends its10-credit reservation. Under the compliant symbol law, expected additional wrong-hold work is(1/4)10=5/2, expected total actual work is75/2, and the worst-case funded total is45. For other admitted guardian exposures the controller retains the same robust reservations and computes actual and expected expenditure from the actual discharge events and the specified joint transcript law. A wrongful retirement additionally consumes its actual successor stock and full transfer work. Release or restoration requires its own authorized funded procedure; an old deadline is not repaired by deleting an alarm.

### E11.4 — Automated cutoff and retained-state quarantine

On an alarm admitted by the stored policy, the controller selects existing preissued denial and escrow operations under their specified triggering conditions for the implicated sub-asset. It does not create a revocation power. It checks and closes every declared qualified incoming supply path; pending inbound commitments are either completed and included in actual retained stock or effectively canceled and fenced before the cutoff certificate is issued. A remaining open or unverified path inhibits that certificate. Readiness to quarantine is therefore a recorded property of the actual boundary, not a request to stop one donor.

At the certified cutoff, let R be the sub-asset's actual accessible work stock and C the remaining finite work yield of explicitly authorized conversion from its retained stock. The system checks the no-external-supply and no-spent-work-reclamation profile. Under that profile its additional paid work is bounded by R+C; at most R+C additional unit-fee ordinary operations are admitted, with discharges and other fees consuming the same allowance. A no-conversion profile sets C=0. The counter is based on actual accounts, not the sub-asset's displayed balance. A new admitted inflow, reclamation operation or changed boundary invalidates the old certificate and requires a new account/profile before further authorization.

Quarantine retains the whole addressed asset, payload, unused stock, spent accounts, permissions, provenance and execution prefix in identified custody. It does not erase occupancy, end a continuing query by removing a row, replenish a purse or claim cessation of unobserved biological activity. The spend bound governs accounted operations in the specified resource boundary. A safe protected lock can persist when a controller funding halt prevents further responsive work; reopening, live monitoring and deadline-sensitive actions require their actual reserves and current guards.

**Prospective stock example.** A cutoff sub-asset with R=5,C=0 cannot fund three ordinary words plus two two-credit discharges, whose work bill is7. One ordinary/discharge event spends3, followed by two ordinary events spending1 each. The retained asset and its remaining obligations persist when available

work reaches0, regardless of a false displayed balance99.

### E11.5 — Configured containment-versus-excision selection

The decision controller compares admissible response plans at a common clock/preparation cut, with the same supplied inventory, service objective, retained-history requirements and event accounting. A containment plan can preserve the old role and perform qualified repairs. An excision plan retires a whole module under standing retirement authority, retains the module and its old accounts, and assigns only genuine unexpired assignable rights to a supplied pre-authorized successor. Cutoff remains enforced so the same implicated incoming route is not admitted at the successor. Each assignment or use checks immutable expiry against the issuer-authorized current clock including elapsed suspension; an unavailable clock inhibits the operation.

For a configured robust-capital comparison, let $D_\tau$ be common already-paid detection work, L common paid preparation work, H the remaining service horizon, $A_X$ the nonfuture protocol and required live-guardian overhead, and $c_X$ the postpreparation spare capacity for response $X \in \{C,R\}$. With sufficient addressed rights/blanks, fixed ordinary and maintenance reserve compartments and the declared full-trigger one-binary-posting future profile, the planner calculates:

$$E_X = D_\tau + L + A_X + H + 2[H - c_X]_+. \tag{G4}$$

$$A_R - A_C < 2([H - c_C]_+ - [H - c_R]_+). \tag{G5}$$

Rule (G5) selects a strictly lower robust-capital retirement plan only after both plans pass actual resource, time, scope and authority checks. A tie uses the stored tie policy. Missing authority, expired rights, insufficient releases or unaffordable setup makes a plan unavailable, not a very large fictitious price. The selected metric can instead compare specified service-success and work/capital coordinates within the configured menu. Actual post-detection occupancy and clock values drive the comparison; detection age is not a universal retirement trigger. A parked, closed containment boundary is not assigned responsive upkeep that the service does not require.

**Prospective threshold inputs.** For a full B-slot old module, empty B-slot successor, H=B, common L=3, $A_C$=0 and $A_R$=38 from a supplied one-handoff configuration, B=19 gives $E_C$=$E_R$=60 before common detection work. B=20 gives63 versus61. Added verification or guardian work is charged at its actual native or common padded cut and can change the selected plan. These are configured service/capital comparisons, not an assertion that retirement dominates every retained-stock or old-root objective.

The succession record is task-minimal relative to a fixed receiving-query contract: retain each needed field unless the receiver can resolve the identical value from an authorized authoritative versioned record. It binds actual successor/source identity, assigned rights and immutable terms, current custody, required provenance and operational prefix. It never substitutes a class code for the actual material. Any optional

E5 compact status code retains E5's fixed independent Bernoulli source, fixed full-trigger policy, status-only/no-feedback and no-additional-conditioned-source restrictions. A record-dependent or biological query keeps its required source and history fields.

Whole-module retirement requires that old local-answer duties have lawfully ended or remain answerable by a qualified retained response path. When continuing good records share an inseparable module, the planner admits only separately authorized and fully priced record transfers or a retained response path satisfying those queries. It does not combine cheap retirement with an invented extraction operation. A previously failed role remains failed in its retained history; succession establishes a forward role.

### E11.6 — Commit, actuation and finite-session integration

The audit and response plans use distinct event keys, payers, retained statistics and actual custody assignments. Each digital resource/permission mutation, completed-event marker and next-prefix value commits atomically as one durable unit; replay returns its stored outcome. Each external command additionally uses the E10.1 execution latch, action-bound observation and terminal-disposition/fencing protocol. No replacement key or affected reservation release is admitted while an old queued or in-flight command remains executable. Permission, calibration and issuer-clock validity are checked at actual actuation; delayed confirmation preserves a completed action without extending permission.

The audit adapter's read-only flag does not rearm its cores or grant them a new assignment transition. A subsequent independent handoff uses a separately supplied, configured, paid and qualified core set and its own authoritative asset catalogue. A transfer already authorized as a suffix of an existing handoff follows that declared suffix, not a newly inferred connector. Reusing any session requires an actually completed, conserving, separately paid and qualified reinitialization; a label change or unused reserve is not that operation. Retained books use cumulative upkeep and cannot recover initial spare capacity for free.

The controller admits only the finite candidate and output envelope for which it reserved records, fresh codes, recipients and guardian upkeep. It records qualification interruption, funding halt and overflow separately. An unprocessed or incomplete audit is not presented as the completed q-sample guarantee. A change to catalogue, fault family, delay, profile or supply boundary triggers requalification before further certified operation.

### E11.7 — Platform embodiments

**Distributed-ledger fraud containment.** A permissioned ledger controller freezes authenticated posting/custody receipts, compares claimed balances, token holders and immutable terms with actual account records, and samples its declared audit window without reading a submitter's hidden program. An alarm invokes preissued submission/supply gates and escrow for the implicated partition. Genuine service rights can move to a qualified successor partition through the priced retirement path; copied descriptors do not create money or authority.

**Storage-fleet anomaly isolation.** A fleet controller monitors retained write, media-identity, custody and maintenance receipts for an identified storage module. The actual account boundary covers permitted I/O work, maintenance stock and outstanding routed commands. It confirms closure or settlement of those commands, retains the module's records and contents, and compares qualified repair with replacement using configured work/capital thresholds. Hardware deterioration and sensor observations remain inputs to the asset-specific acceptance profile.

**Bioprocess contamination or deviation response.** An asset-records controller monitors authenticated material identity, QC/deviation and custody evidence for a managed organism, culture, population or preserved-material sub-asset. A configured QC failure or statistical alarm can activate preissued supply cutoff and physical-plus-record quarantine. Assay aliquots, wrong-hold costs, environmental support, maintenance work and actuator confirmation are accounted separately. The cutoff allowance describes the controlled resource account, not a biological growth rate or a prediction of viability. Excision uses the B2 production-role succession path, actual retained material, genuine unexpired permissions and a supplied pre-authorized successor. The B1–B5 acceptance, bounded-response, guardian, actuation and paid-initialization requirements remain in force for the selected operations. Material is routed to the managed process rather than a patient.

### E11.8 — Sampling component note

R's stats documentation, *The Hypergeometric Distribution*, identifies without-replacement probability, distribution and random-generation interfaces: https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Hypergeometric.html. The controller implements the finite counts and qualified selector interface stated in E11.2; its actual evidence, private selection, finite resources and response authority are supplied through the specified host interfaces.

## PROPHETIC EXAMPLES

**P1 — Pruning admission.** In a prospective configuration, the input sequence contains one funded qualified discharge, one identical request lacking the addressed release, and one lacking ordinary funds. The controller is configured to perform only the qualified operation, retain all attempted prefixes, and distinguish Q_h/F_h from an actual subsequent O_h. A release for a different obligation is rejected.

**P2 — Frontier boundary.** A prospective configuration uses the sixteen binary arrival words for E3's B=3,z=2,H=4 case. With ten qualified credits the controller is configured to sustain all words; with eight credits and the fixed allocation it stops on exactly the all-arrival word. Replacing one valid release with an irrelevant one causes the controller to withhold the robust certificate even with ten credits.

**P3 — Transfer safety.** In a prospective configuration, proposals alter only nonce, purse amount, recipient or expiry. The interface is configured to reject each mismatch against actual registry state. A halt immediately after assignment leaves READY false and actual assets moved once; resuming the receipt suffix does not move them again. A duplicate proposal does not restore old spending authority. In an N=2

case, a failed first attempt followed by the valid current nonce 2 is compared with m_2 and is eligible for acceptance upon satisfaction of the remaining guards. Crash cuts of each keyed update expose either its complete pre-state or complete post-state, including the resource mutation and marker together. With planned cut 10 and expiry 10, recovery at issuer-authorized time 11 refuses an outstanding expiry-sensitive assignment; an already completed assignment is retained without repetition. Unavailable authoritative time inhibits qualification.

**P4 — Guardian and session boundary.** A prospective configuration supplies four all-IDLE core exposures with only three ordinary credits. The core is configured to report a funding halt at the fourth request while occupancy remains zero. With c_i=1,T=4, ten qualified credits per core reserve resources for every exposure word. After completion, a begin request lacking a conserving paid reinitialization receipt remains inhibited, regardless of unused nominal capital or a fresh session label.

These operating examples use the specified transaction substrate, fault domain, resource reservations and authoritative clock. The completed-session interlock requires the separately specified and actually completed paid initialization before subsequent operation.

## CLAIMS

1. A computer-implemented method of controlling maintenance of a finite-register service, comprising: storing a register-slot inventory, actual credit balances, obligation-specific release permissions, recipient identities and a no-reclamation, no-unrecorded-outside-input operation profile; receiving a proposed schedule having H_total ordinary words, D two-credit discharges and other distinct fees F; checking $H_total+2D+F \le V_0+(T_0-T_min)/\varepsilon_W$, where $V_0$ is opening work stock, $T_0$ and T_min are opening and minimum retained tariff of a specified convertible account, and ε_W is positive tariff per credit; separately checking actual timely delivery, release, exclusive custody and recipient/headroom qualifications for each proposed operation, without treating the tariff bound as an implemented converter; reserving actual resources for an admitted schedule; executing admitted operations against the register while debiting each paid event once and retaining displaced obligations; and recording funding halt, capacity overflow and qualification interruption as distinct service outcomes at their actual operation prefixes; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.
2. The method of claim 1, wherein the operation profile orders ordinary payment, top-up, maintenance, arrival encounter, clock advancement and receipt, and the recorded outcome identifies the first obstructing stage, with qualification interruption preceding funding halt when both occur at the same stage.
3. The method of claim 1, wherein a maintenance top-up into a valid local purse of balance w and capacity W transfers $\min(\max(0,\text{required}-w),\text{accessible_donor},W-w)$, and an invalid purse state is rejected without being clipped into validity.
4. The method of claim 1, wherein an obligation-specific release identifies the actual occupied slot and a distinct qualified recipient, and payload canonicalization that does not free the slot is charged as another fee without incrementing D.
5. The method of claim 1, further comprising retaining stage and paid-event identifiers after interruption and resuming only an outstanding suffix, while recording a repeated attempt as a separately funded attempt rather than repeating a completed debit or transfer; wherein returning the stored outcome for a completed event key does not repeat its atomically committed debit, move or prefix advancement.
6. The method of claim 1, wherein the finite-register service is a permissioned replicated storage service, and log-pruning operations are routed through a designated ordered executor that checks retention permissions against authoritative registry state before releasing an occupied service slot.

7. A finite-register maintenance system comprising: an addressable obligation store of bounded capacity; a registry of actual credit accounts, release permissions, receiver availability and exclusive custody; a finite spent-resource store; and a processor coupled to the stores and configured to calculate a finite-fuel envelope $V_0+(T_0-T_min)/\varepsilon_W$ under a stored no-reclamation and no-unrecorded-input profile, compare planned ordinary words plus twice planned discharges plus other distinct fees with the envelope, separately verify actual conversion/delivery and obligation-specific qualification, reserve and debit actual resources for admitted maintenance, retain discharged obligations in identified recipients, and inhibit an unavailable next operation while separately reporting funding, qualification and overflow outcomes; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

8. The system of claim 7, wherein the processor refuses a debit when spent-resource or retained-record headroom is unavailable and preserves the actual completed prefix with a completion receipt reported only for a completed operation.

9. A non-transitory computer-readable medium storing instructions that, when executed by processors coupled to a bounded obligation store and authoritative resource registry, cause the processors to: receive a proposed maintenance schedule; compute ordinary-word, two-credit-discharge and distinct-fee totals; compare their sum against $V_0+(T_0-T_min)/\varepsilon_W$ for a selected no-reclamation, no-unrecorded-input profile; verify actual timely resources, release, custody and recipient qualifications independently of that bound; reserve and execute only admitted maintenance operations with one debit per paid event; and retain distinct funding-halt, qualification-interruption and overflow records at the actual operation prefix; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

10. The medium of claim 9, wherein the instructions further prevent a changed operation family, outside-input rule or reclamation rule from reusing an admission certificate calculated for the unchanged finite-fuel profile.

11. A computer-implemented method of provisioning a bounded-register service for a horizon H, comprising: receiving capacity $B\geq1$, occupancy z, initial spare capacity $c=B-z$, integer allocable credits E and a profile permitting at most one binary arrival per funded word and qualified maintenance before the unrevealed arrival; calculating a robust reservation $H+2[H-c]_+$; identifying H actual ordinary credits and $[H-c]_+$ qualified two-credit maintenance services with timely routes, addressed releases, distinct blanks and required record/spent headroom; rejecting a robust plan

when either the credit reservation or a required qualification is unavailable; and otherwise assigning the identified resources to fixed horizon compartments and directing a maintenance controller to discharge a full register before an arrival while qualified service remains.

12. The method of claim 11, further comprising calculating effective spare capacity $\ell=c+m$ with $m=\min(\lfloor M/2 \rfloor, Y, R)$, where M is accessible maintenance credit, Y is usable distinct blank count and R is usable universally applicable release count, and reducing m for any additional service-opportunity limit.

13. The method of claim 11, further comprising, for a declared IID Bernoulli arrival parameter $0<a<1$ and nonbinding stock caps, reporting zero horizon survival when $E<H$ and otherwise computing $\Sigma[j=0..\min(c+\text{floor}((E-H)/2),H)]\ \text{binom}(H,j)a^j(1-a)^{(H-j)}$, labeled as the qualified fixed-horizon allocation result.

14. The method of claim 13, wherein the method refuses to apply the IID calculation to a declared correlated arrival process and retains the actual source-law identifier with the allocation.

15. The method of claim 11, further comprising selecting a robust horizon equal to E when $E\le c$ and otherwise $\text{floor}((E+2c)/3)$, verifying the corresponding qualifications, and issuing reservations for the selected horizon without combining separately optimized horizons into one survival process.

16. The method of claim 11, wherein the resource allocation retains an odd credit left after allocation of maintenance pairs without treating that credit as a complete two-credit service.

17. A horizon-provisioning system comprising: a capacity and occupancy interface providing register capacity B, occupancy z and a requested integer service horizon H; a registry of actual ordinary and maintenance credits, release scopes, recipient stocks and delivery routes; a processor configured to verify integer $B\ge1$, $0\le z\le B$, $c=B-z$ and $H\ge0$; enforce a fixed profile with at most one binary arrival per funded word, qualified full-trigger maintenance before the unrevealed arrival, an accessible local maintenance purse admitting a two-credit service, and fixed ordinary and maintenance compartments; and calculate $H+2[H-c]_+$ only within that profile, verify $[H-c]_+$ actually qualified maintenance services, reserve H ordinary credits and twice that service count as maintenance credits in fixed compartments, and refuse a robust certificate on missing qualification despite a sufficient nominal balance; and a maintenance-control interface configured to issue the resulting full-trigger discharge schedule to the bounded register.

18. The system of claim 17, wherein the processor outputs separate fields for reserved capital, expected expenditure, service probability and qualification state so that a lower expected expenditure due to interrupted service is not reported as a robust funding certificate.

19. A non-transitory computer-readable medium storing instructions that, when executed by a processor coupled to a bounded-register controller, cause the processor to: receive register capacity B, occupancy z, a requested integer horizon H, spare capacity c and actual allocable resources; verify integer $B\ge1$, $0\le z\le B$, $c=B-z$ and $H\ge0$; enforce a fixed profile with at most one binary arrival per funded word, qualified full-trigger maintenance before the unrevealed arrival, an accessible

local maintenance purse admitting a two-credit service, and fixed ordinary and maintenance compartments; and calculate $H+2[H-c]_+$ only within that profile; validate timely ordinary credits, two-credit maintenance pairs, addressed releases and distinct recipient stock; reserve those resources before the opening cut; and command full-trigger maintenance only within the verified profile, withholding the robust service certificate when a required qualification fails.

20. The medium of claim 19, wherein the instructions further generate an asset-fleet maintenance allocation only under a stored mapping of actual maintenance requirements, work prices, restoration capacity, assignment permissions and service windows to the selected register profile.

21. A computer-implemented method of succession in a finite-register service, comprising: receiving an identified service-role contract, a supplied successor module, a selected typed transfer law and a prospectively fixed comparison cut; identifying actual records, purses, capacity titles and exclusive rights that the law permits to move and actual old records that must remain; checking the successor's pre-authorized eligibility, original right scopes, remaining uses and immutable expiry against the issuer-authorized current clock value at each expiry-sensitive operation, and treating the prospectively fixed comparison cut only as the reserved schedule; reserving the identified resources and distinct operation fees; executing the selected guarded transfer to the successor while preserving original right terms, actual retained old records and the completed operation prefix; recording whether the old root is retired and which service role continues; and publishing successor readiness only after completion of the paid transfer and its final receipt; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; further comprising maintaining separate ordinary-operation, discharge and transfer-fee reservations under a stored mode-specific operation table; distinguishing movement of occupied records from a two-credit discharge that actually releases an active slot; and refusing a claimed fresh-role opening unless preissued whole-module retirement authority, exclusive successor rights, retained old-module custody and the paid opening resources are all present; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

22. The method of claim 21, wherein the selected law retires a whole occupied module under preissued retirement authority, retains its payloads in old-module custody, activates a separately supplied fresh module and assigns an exclusive continuing role/right bundle to its already eligible root without promising old payload queries that were not separately retained or lawfully released.

23. The method of claim 21, wherein an absolute expiry is unchanged by assignment and a successor-contingent grant is accepted only if issued before the transfer under terms already authorizing the successor; wherein each expiry-sensitive operation uses the issuer-authorized current clock required by the method rather than a paused local step index.

24. The method of claim 21, further comprising, only for a fixed qualified profile with independent Bernoulli(a) future arrivals for a fixed $0<a<1$, a fixed full-trigger maintenance policy, and observations confined to current and future live status without payload, cause, policy-selection or archive-feedback queries: determining at the post-payment advanced cut a remaining integer horizon H, actually qualified ordinary words o and effective spare capacity $\ell$; encoding stopped status or a live pair (j,k), where $j=\min(o,H)$, $k=\min(\ell,j)$ and $0\le k\le j\le H$; binding the encoding to the fixed profile, a, advanced cut and actually supplied future resources; and retaining payload, authority and resumption records separately, wherein the status encoding is not substituted for conditioned source state in a lineage transfer outside that profile.

25. The method of claim 21, wherein the selected law is one of: a record-only transfer retaining the pending obligations without an implied purse; a full-transferable-custody transfer moving named records, existing funds and permitted rights without treating moved occupancy as discharge; or a lineage-root reassignment preserving actual conditioned source state, ancestry and existing finite funding without redrawing an independent source.

26. The method of claim 21, further comprising latching a completed-session state and inhibiting a subsequent session unless an authoritative receipt identifies an actually completed conserving and paid reinitialization, or separately supplied and paid fresh initialized cores, together with retained old-state custody, current permissions, stock and exposure qualifications.

27. A verified-succession system comprising: a roster of old and supplied successor modules; a registry of exclusive rights with issuer, scope, permitted successors, uses and immutable expiry; a typed transfer controller configured to select which named records and assets move under a record-only, rights-only fresh-role, full-transferable-custody or lineage-root law, verify current eligibility and terms at an advanced cut, reserve distinct payment resources, execute guarded custody and role changes while retaining old records and operation prefixes, and issue readiness only on a final completion receipt; and a session interlock configured to prevent a completed session from being treated as a fresh opening without comparison of authoritative actual opening state with a separately specified and actually completed paid initialization, or with separately supplied paid fresh initialized cores, including retained old-state custody, current permissions, fresh reference stock and the applicable exposure envelope; a receipt alone does not execute or supply the transition; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the

old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; the typed transfer controller further configured to perform operations comprising: maintaining separate ordinary-operation, discharge and transfer-fee reservations under a stored mode-specific operation table; distinguishing movement of occupied records from a two-credit discharge that actually releases an active slot; and refusing a claimed fresh-role opening unless preissued whole-module retirement authority, exclusive successor rights, retained old-module custody and the paid opening resources are all present; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

28. The system of claim 27, further comprising a comparison processor configured to compare migration and deferred repair at aligned preparation duration and actual resource access, retain separate work, reserved-capital, survival and residual-stock outputs, and identify expiry timing or changed rights access rather than erasure as the basis for a reported advantage.

29. A non-transitory computer-readable medium storing instructions that, when executed by processors coupled to a module roster and custody registry, cause the processors to: select a typed succession law for a declared service contract; verify actual transferred assets and preissued successor eligibility at a stated cut; preserve immutable expiry and exclusive right uses; perform a paid guarded transfer while retaining old payload custody and intermediate transfer-prefix state; distinguish retired-root status from continuing-role status; and publish readiness only after the final completion receipt, without automatically clearing the completed session for reuse; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; the instructions further causing the processors to perform operations comprising: maintaining separate ordinary-operation, discharge and transfer-fee reservations under a stored mode-specific operation table; distinguishing movement of occupied records from a two-credit discharge that actually releases an active slot; and refusing a claimed fresh-role opening unless preissued whole-module retirement authority, exclusive successor rights, retained old-module custody and the paid opening resources are all present; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its

durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

30. The medium of claim 29, wherein the instructions further, only under a fixed qualified profile with independent Bernoulli(a) future arrivals for a fixed $0<a<1$, a fixed full-trigger maintenance policy and status-only observations without archive feedback: determine remaining integer horizon H, qualified ordinary words o and effective spare capacity $\ell$ at the post-payment advanced cut; set $j=\min(o,H)$ and $k=\min(\ell,j)$; encode stopped status as 0 and a live pair as $1+j(j+1)/2+k$; bind the encoding to that profile, a, cut and future-resource declaration; and omit an additional synopsis only when actually supplied side information already determines the same class, while retaining payload, permission, conditioned-source and resumption records outside that status family separately.

31. A computer-implemented escrow transfer method comprising: reserving an actual identified asset bundle under exclusive permission control; generating a retained canonical authoritative session manifest from actual-state fields; scheduling a finite number N of separately funded proposal slots; receiving a proposal descriptor through a lane whose admitted faults affect descriptors rather than the reserved assets or authoritative permission controls; deriving the expected complete descriptor for each attempt from the retained authoritative session manifest and the current protected slot nonce, and comparing every received field with that attempt-specific descriptor; checking actual locks, amounts, terms, receiver reservation and final-cut qualification; retaining each accepted or rejected attempt and its charges; after a valid proposal, rechecking the actual guards and executing the selected guarded asset assignment once; and withholding readiness until its final paid receipt, wherein a received descriptor does not itself confer credit, custody or spending authority; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; wherein the method is executed by verifier, escrow and commit controllers each having a separately funded finite upkeep book; further comprising reserving separate protocol-fee and future-service compartments through a prospectively fixed final cut, paying every complete rejected attempt at the same declared full-verification price as an accepted attempt, executing paid IDLE in unused scheduled slots after acceptance, and gating each transfer stage on the separately funded finite upkeep books of the verifier, escrow and commit controllers; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each

executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

32. The method of claim 31, wherein each complete active proposal has ordinary price 1 and extra price d=(e+k_c+v)t+g for t cells, including full-verification charge vt+g, and a complete rejected proposal pays the same scheduled price as an accepted proposal.

33. The method of claim 32, further comprising reserving $K_0+\kappa N$ for configuration of supplied finite apparatus and nonreused kits, keeping this reservation distinct from raw-factor production, transferred funds and future-service reserves, and retaining unused kits as inventory.

34. The method of claim 33, wherein N=b+1 covers at most b failed candidates before a valid one and the method computes additional capital b(d+κ) relative to a one-attempt branch at the same paid padded cut, while separately charging changed duration and refusing to cure an expired right by adding that capital.

35. The method of claim 31, wherein a current valid first proposal is accepted, an earlier nonce or committed-session replay cannot cause another assignment, remaining slots after acceptance execute paid IDLE, and no valid proposal by N produces an abort without refunding spent fees or renewing expired rights; wherein the once-only assignment and its completed-event marker share the atomic durable update, and a replay returns its stored outcome.

36. The method of claim 31, further comprising recording a transfer obstruction selected from unavailable exclusive asset, absent or withdrawn receiver reservation, detected descriptor/authority inconsistency, and expired required right; refusing completion on the obstruction; and withholding an authenticity guarantee when the sole authoritative view is outside the trusted actual-state boundary.

37. A transfer-boundary system comprising: an exclusive actual-asset escrow registry; a retained authoritative session-manifest store; an escrow controller managing the registry; finite proposal, quarantine and receipt buffers; a full-field verifier operating under a descriptor-lane fault profile and configured for deriving the expected complete descriptor for each attempt from the retained authoritative session manifest and the current protected slot nonce, and comparing every received field with that attempt-specific descriptor; separate protocol-fee and future-service accounts; and a commit controller configured to recheck actual locks, balances, receiver reservation and original right terms after descriptor acceptance, execute a selected guarded custody assignment once, preserve already executed prefixes on interruption, and issue readiness only after the final receipt, without treating a proposal or acceptance latch as a transferable asset; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation

certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; the verifier, escrow and commit controllers each having a separately funded finite upkeep book and the system further configured to perform operations comprising: reserving separate protocol-fee and future-service compartments through a prospectively fixed final cut, paying every complete rejected attempt at the same declared full-verification price as an accepted attempt, executing paid IDLE in unused scheduled slots after acceptance, and gating each transfer stage on the separately funded finite upkeep books of the verifier, escrow and commit controllers; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

38. The system of claim 37, wherein an abort returns old-holder access only when the original terms still permit it, and the finite buffers and spent stores inhibit an operation when their actual headroom is unavailable.

39. A non-transitory computer-readable medium storing instructions that, when executed by processors coupled to an authoritative escrow registry and finite communication buffers, cause the processors to: reserve exclusive existing assets; retain an authoritative session manifest generated from actual-state fields; generate canonical session-bound proposals; derive the expected complete descriptor for each attempt from the retained authoritative session manifest and the current protected slot nonce; compare every received field with that attempt-specific descriptor and check current actual state; pay and retain each admitted attempt within a fixed cap; preserve locked balances separately from attempt fees; recheck qualification at commitment; assign the specified assets once; and retain aborts, assignments made through an incomplete transfer prefix and withheld-readiness states without duplicate spending or implicit rollback; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; the instructions further implementing verifier, escrow and commit controllers each having a separately funded finite upkeep book and causing the processors to perform operations comprising: reserving separate protocol-fee and future-service compartments through a prospectively fixed final cut, paying every complete rejected attempt at the same declared full-verification price as an accepted attempt, executing paid IDLE in unused scheduled slots after acceptance, and gating each transfer stage on the separately funded finite upkeep books of the verifier, escrow and commit controllers; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each

executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

40. The medium of claim 39, wherein the instructions further check that the declared finite cell alphabet and cell count encode the entire capsule catalog injectively and reject an out-of-catalog or noncanonical proposal without substituting a finite hash for full-state equality.

41. A method of operating guardian machines for a finite-register transfer, comprising: supplying verifier, escrow and commit machines each indexed by i and having a finite obligation book of integer capacity B_i, separate ordinary and maintenance accounts, actual release/recipient stock and retained-code/spent headroom; initializing actual escrow, a retained authoritative session manifest and clear session prefixes through paid finite field-read and permission-confined dispatch operations; selecting a preparation exposure horizon T and each machine's integer initial spare upkeep capacity $1\le c_i\le B_i$; allocating to each machine $T+2[T-c_i]_+$ credits with corresponding timely qualified service resources under a one-binary-exposure-per-word profile; executing each machine's upkeep before its corresponding transfer stage; deriving the expected complete descriptor for each attempt from the retained authoritative session manifest and the current protected slot nonce, and comparing every received field with that attempt-specific descriptor; performing full verification, exclusive access control and once-only guarded assignment through the initialized machines; and inhibiting the next transfer action when a required machine has a funding, qualification or overflow interruption, while retaining its actual custody and operation prefix; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

42. The method of claim 41, wherein the supplied field-read/dispatch operations read named actual finite fields into blank pointers, dispatch a bounded program table while retaining old labels, and enforce the same issuer and lock permissions on every admitted operand route, with proposal faults excluded from those reads, tables and permission controls.

43. The method of claim 41, wherein the verifier derives the expected complete descriptor for each attempt from the retained authoritative session manifest and the current protected slot nonce, compares every received field with that attempt-specific descriptor, checks actual asset terms and expected post-fee balances, retains displaced flags and pointers, and latches acceptance only after all tests, without using a status-only succession quotient to replace the complete reference.

44. The method of claim 41, wherein the escrow machine denies source access to reserved operands, uses an upkeep donor separate from locked beneficiary funds, and changes access only by a matching guarded commit or abort suffix.

45. The method of claim 41, wherein the commit machine retains distinct assigned and receipted phases, uses a supplied atomic assignment of listed coordinates, withholds READY after assignment until completion of the final receipt, and resumes only that remaining suffix after an interruption; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; wherein the atomic assignment includes its durable completed-event marker and next-prefix value, while the final receipt is a later keyed atomic update.

46. The method of claim 41, further comprising accounting once for paid setup J and separate upkeep $3T+2\Sigma_i[T-c_i]_+$ in addition to already priced interface events, and identifying and removing any duplicate charge for an event deliberately shared by an alternative implementation.

47. A guardian-core apparatus for a selected finite integer exposure horizon $T\geq 0$ comprising: a verifier with finite retained authoritative session-manifest, current protected slot-nonce, scan, pointer, acceptance and processed-session registers; an escrow controller with actual-asset owner, lock, session, recipient and immutable-term registers; a commit controller with guarded transfer-prefix, assigned and receipted states; a permission-confined finite field-read/dispatch substrate and a guarded atomic assignment operation; and, for each controller i, a separate upkeep book of integer capacity $B_i$, actual ordinary and maintenance reserves and qualified recipients, the apparatus being configured to store and enforce a finite exposure profile with at most one binary posting per funded upkeep word, full-trigger upkeep before the unrevealed posting, integer initial spare capacity $1\leq c_i\leq B_i$, fixed ordinary and maintenance compartments, and timely qualified releases and distinct recipients; to admit a T-epoch transfer only after reserving $T+2[T-c_i]_+$ credits for each core within that profile; and to withhold that robust certificate when the exposure profile or any required qualification changes; and to block a transfer stage upon a required core's interruption without granting that core additional asset authority; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or

use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; the verifier further configured for deriving the expected complete descriptor for each attempt from the retained authoritative session manifest and the current protected slot nonce, and comparing every received field with that attempt-specific descriptor; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

48. The apparatus of claim 47, wherein a finite sink retains actual spent credits, discharged obligations and displaced control records in separately identified recipients and inhibits payment before headroom exhaustion, without treating a spent counter as replenished work or a newly blank recipient.

49. A non-transitory computer-readable medium storing instructions that, when executed by processors coupled to finite guardian books indexed by i with integer capacities B_i and authoritative permission-controlled registers, cause the processors to: perform paid setup from actual asset fields and retain an authoritative session manifest; select a finite integer exposure horizon $T\geq 0$; store and enforce a finite profile with at most one binary posting per funded upkeep word, full-trigger upkeep before the unrevealed posting, integer initial spare capacity $1\leq c_i\leq B_i$ and fixed ordinary and maintenance compartments; allocate separate qualified worst-case upkeep reserves $T+2[T-c_i]_+$ to verifier, escrow and commit programs only within that profile; execute their upkeep before transfer stages; derive the expected complete descriptor for each attempt from the retained authoritative session manifest and the current protected slot nonce; compare every received field with that attempt-specific descriptor; perform full comparisons, exclusive permission checks and once-only guarded assignment; distinguish each program's funding halt from overflow; and preserve actual prefixes and withhold readiness upon an incomplete required stage; wherein each applicable right specifies its expiry clock domain and issuer-authorized clock source, and each expiry-sensitive assignment or use is permitted only after checking immutable expiry against that source's authoritative current value including elapsed suspension in its domain; an unavailable clock value qualification-inhibits the operation, the planned cut is only a reserved schedule, and an interruption outside the declared delay/exposure envelope invalidates the old completion and robust-reservation certificate and requires rechecking remaining rights, actual time, headroom, exposure and funding before the outstanding suffix; wherein each fine-stage command has an immutable event key comprising profile_version, session, attempt, stage and operation_index; and wherein each executed fine-stage mutation and its durable completed-event

marker and next-prefix value are committed as one atomic state-machine update, and recovery returns the stored outcome for an already completed event key without repeating its resource mutation.

50. The medium of claim 49, wherein the instructions further retain a completed-session latch, calculate the necessary worst-case upkeep reservation $3nT+2\Sigma_i[nT-c_i]_+$ for a proposed n-session schedule within that same exposure profile using the same retained upkeep books, while recording actual expenditure separately, and refuse renewed initial spare capacity or a next session without an actually completed, conserving, separately paid and qualified reinitialization or supplied-new-core receipt.

51. The method of claim 1, wherein the finite-register service controls a managed biological asset comprising one or more organisms, a culture, a population, or preserved biological material; the register-slot inventory addresses an obligation store associated with a record of passage or generation count and a senescence or quality-control state class; an admitted operation is a maintenance intervention against identified actual stock; qualification includes the intervention-specific release and chain of custody of the stock and receiving asset; a discharge closes an addressed protocol obligation only after the specified acceptance evidence while retaining passage history and original records; and the method separately records credit-exhaustion funding halt, register overflow, an observed asset damage-saturation flag and qualification interruption, without treating a record update as a change of biological state.

52. The system of claim 17, further comprising a biological production asset, an asset-records system storing the asset identity, passage or generation history and quality-control class, and an actuation interface, wherein B and z are the capacity and occupancy of a protocol-obligation register associated with the asset, H counts controller service words rather than cell divisions, and the processor reserves identified maintenance stock, addressed releases, handling capacity and recipients for the fixed-horizon schedule and withholds the robust certificate when the protocol-posting profile or an actual supply qualification fails.

53. The medium of claim 19, wherein the instructions cause the processor to provision a biological-asset management schedule using an asset information record binding asset identity, passage or generation history, quality-control class and actual intervention inventories; to apply the fixed binary-arrival profile to protocol-obligation postings; to retain biological observations separately; and to require a uniquely keyed confirmation of the physical intervention from a receiver adapter before discharging an obligation addressed by that intervention, while preserving the ordinary/maintenance reservations and withholding certification on a changed exposure profile or missing qualification; wherein the adapter comprises an actuator boundary; wherein the adapter refuses a replacement operation key and release of affected reservations until the actuator boundary durably identifies the prior action as completed or canceled with no executable command remaining, rejects a delayed command from a retired fencing generation, and binds its confirmation evidence to the operation key, generation and reserved physical action.

54. The method of claim 21, wherein an old module of the finite-register service is a biological asset occupying a named production role, the supplied successor is a biological asset established from an identified preserved biological succession asset comprising a whole organism, a culture, a biological line, a propagule, or biological material sufficient to establish a successor organism, culture or biological line, and the selected transfer retires the old asset from that role while assigning an existing assignable license, consent-based permission or quota to the pre-authorized successor; further comprising preserving the old asset records and remaining material in identified custody, verifying current withdrawal or revocation conditions specified by the original issuer terms, and withholding successor readiness until actual establishment, stock consumption, paid initialization and final completion are confirmed, without extending the original expiry or resetting the old provenance history.

55. The system of claim 27, wherein the roster comprises an old biological asset, a pre-authorized successor, and a preserved biological succession asset comprising a whole organism, a culture, a biological line, a propagule, or biological material sufficient to establish a successor organism, culture or biological line; and wherein the typed transfer controller is configured to form a minimal-transfer snapshot comprising the actual identified succession asset and a status record bound to its custody; to select the status fields from a fixed receiving-query contract requiring succession-asset and lineage identity, provenance-history reference, current quality-control and preservation evidence, material quantity, applicable rights and operating prefix; to omit a required field only when the receiver can uniquely resolve the identical value from an authorized authoritative versioned record; and to retain other lineage, permission and history records separately while requiring actually completed paid successor establishment before readiness.

56. The medium of claim 30, wherein an actual material factor supplied for succession is an identified preserved biological succession asset comprising a whole organism, a culture, a biological line, a propagule, or biological material sufficient to establish a successor organism, culture or biological line, the succession asset being accompanied by a bound status record, the fixed independent-arrival and status-only profile applies to the controller protocol-obligation stream, and the instructions further compare the actual succession-asset identifier and current custody and preservation evidence with the versioned receiving-query record, bind the class code to the verified succession asset and declared receiving-query contract only upon that comparison, inhibit succession-readiness on a failed or unavailable comparison, and omit a redundant synopsis only when the transferred side information already determines that same class, retain biological lineage, quality-control, custody, permission and resumption records separately, and refuse to substitute the class code for the actual succession asset or for conditioned biological-source state.

57. The method of claim 31, wherein the reserved asset bundle comprises a biological sample or preserved biological material comprising a reserved specimen, a uniquely identified aliquot, or an identified organism or portion thereof; wherein that biological sample or preserved biological material constitutes the sample and acceptance additionally requires current identity and quality-control evidence bound to the actual sample, a stored acceptance rule, authorized test issuer,

handling envelope and receiver capacity; wherein test aliquots, consumables and verification charges are reserved and consumed separately from the transferred sample; and wherein exclusive-material loss, receiver or contamination/quality-control refusal, descriptor/identity inconsistency and elapsed consent or license are separately recorded obstructions, with a failed sample qualification retaining quarantine rather than being cleared by descriptor replay.

58. The system of claim 37, further comprising a biological-material custody interface, a sample-identity and quality-control evidence registry and a journaled physical-transfer adapter, wherein the adapter durably latches execution of the corresponding immutable event key before issuing its actuator command, refuses to issue that key again, and returns a confirmation only upon the prescribed identity, position and quantity evidence; wherein an uncertain physical outcome locks the affected sample and inventory in an indeterminate state for authorized reconciliation; and wherein the commit controller treats authorization and physical confirmation as separate keyed atomic updates and withholds final readiness until actual receiving custody and all current guards are confirmed; wherein the adapter comprises an actuator boundary; wherein the adapter refuses a replacement operation key and release of affected reservations until the actuator boundary durably identifies the prior action as completed or canceled with no executable command remaining, rejects a delayed command from a retired fencing generation, and binds its confirmation evidence to the operation key, generation and reserved physical action.

59. The medium of claim 39, wherein the instructions manage escrow of biological samples using an asset information record, reserve unique parent-linked assay aliquots and their actual quantities, apply stored identity and quality-control acceptance rules, derive each sample descriptor from the retained authoritative manifest and current protected nonce, and retain consumed test stock and full scheduled verification charges on rejection; and wherein the instructions distinguish a sample qualification failure from a descriptor-lane error, preserve separate loss, refusal, inconsistency and expiry records, and require confirmed actual custody through a uniquely keyed receiver adapter before the final readiness receipt; wherein the adapter comprises an actuator boundary; wherein the adapter refuses a replacement operation key and release of affected reservations until the actuator boundary durably identifies the prior action as completed or canceled with no executable command remaining, rejects a delayed command from a retired fencing generation, and binds its confirmation evidence to the operation key, generation and reserved physical action.

60. The method of claim 1, wherein the service controls a supplied engineered organism or culture having a proliferation- or viability-supporting function dependent under a stored component profile on an externally supplied factor; wherein actual factor stock is held in a custody-controlled reservoir and each factor credit is a uniquely issued authorization bound to a reserved lot quantity, culture identity, maximum dispense quantity, valid interval and operation key; and further comprising checking current expiry, actual stock, release, calibrated delivery route and observed headroom under the stored component profile before authorizing a metered dispense, consuming the authorization once by the atomic state-machine update, confirming delivery separately through a journaled receiver adapter, and inhibiting further dispensing on credit exhaustion or failed

admission, while retaining observed culture state separately from the command-halt state; wherein the stored component profile identifies a monitored process variable, common units, a finite response interval, an upper-increment table for each admitted dispense quantity and a residual-response upper bound, and admission compares the sum of those two bounds with the available headroom of that variable while retaining outstanding delivery reservations; wherein the adapter comprises an actuator boundary; wherein the adapter refuses a replacement operation key and release of affected reservations until the actuator boundary durably identifies the prior action as completed or canceled with no executable command remaining, rejects a delayed command from a retired fencing generation, and binds its confirmation evidence to the operation key, generation and reserved physical action.

61. The system of claim 7, further comprising a supplied nonpathogenic engineered organism or culture with an externally dependent proliferation- or viability-supporting component, a custody-controlled inducer or dependency-factor reservoir, a metered dispenser, an organism-state observation interface and a durable actuator adapter; wherein the processor binds uniquely issued factor credits to actual reserved quantities and current permissions, refuses credit creation from culture replication, label changes or descriptor copies, gates each dispense on qualified supply and admissible headroom under a stored component profile, retains an execution latch that prevents a second actuator command for the same key, locks uncertain delivery for reconciliation, and requires an actually completed paid culture-and-controller initialization before a subsequent session; wherein the stored component profile identifies a monitored process variable, common units, a finite response interval, an upper-increment table for each admitted dispense quantity and a residual-response upper bound, and admission compares the sum of those two bounds with the available headroom of that variable while retaining outstanding delivery reservations; wherein the adapter comprises an actuator boundary; wherein the adapter refuses a replacement operation key and release of affected reservations until the actuator boundary durably identifies the prior action as completed or canceled with no executable command remaining, rejects a delayed command from a retired fencing generation, and binds its confirmation evidence to the operation key, generation and reserved physical action.

62. The medium of claim 9, wherein the instructions control factor dispensing to a supplied engineered organism or culture having a supplied external-factor-dependent proliferation or viability function, separately track issued authorizations, reserved material, confirmed delivered material and spent factor credits, and cause the processors to refuse a new dispense when a component-profile upper bound for the requested dispense and a separately reserved upper bound for residual response, expressed in the units of the same monitored process variable over the same declared finite response interval, exceed the available headroom of that variable, and to refuse admission when that bounded profile or its required observation is unavailable or invalid; to return the stored outcome for a repeated operation key without another dispense; to record uncertain delivery as a qualification interruption; and to preserve spent credits, culture observations and old-session records through re-entry until actual paid initialization, reconciled stocks and current permissions

are verified, the dispensing being to the culture process rather than to a patient; wherein the instructions further implement a receiver adapter for the dispensing; wherein the adapter comprises an actuator boundary; wherein the adapter refuses a replacement operation key and release of affected reservations until the actuator boundary durably identifies the prior action as completed or canceled with no executable command remaining, rejects a delayed command from a retired fencing generation, and binds its confirmation evidence to the operation key, generation and reserved physical action.

63. The method of claim 41, wherein the verifier, escrow and commit machines operate as automation subsystems of a biological operations facility, each upkeep book records calibration, inspection or scheduled controller-maintenance obligations or other scheduled upkeep obligations, and the method enforces full-trigger upkeep before the unrevealed binary posting with fixed ordinary and maintenance compartments; further comprising checking actual instrument-calibration releases and current validity before sample transfer, culture-factor dispensing or successor-readiness stages, charging the cores separately from locked sample or beneficiary assets, and inhibiting a dependent stage on a core funding, qualification or overflow interruption while retaining its custody and execution prefix.

64. The apparatus of claim 47, wherein the verifier is coupled to sample-identity, quality-control and calibration records, the escrow controller is coupled to biological-asset locations or supply lots, and the commit controller is coupled to a command-and-confirmation adapter; wherein the apparatus requires actual completed calibration evidence and issuer-clock validity in addition to the qualified worst-case upkeep reservation for each core, records actual upkeep expenditure separately, and invalidates a dependent transfer or dispense stage upon expired calibration or a changed exposure profile without using locked beneficiary resources to fund the interrupted core; wherein the adapter comprises an actuator boundary; wherein the adapter refuses a replacement operation key and release of affected reservations until the actuator boundary durably identifies the prior action as completed or canceled with no executable command remaining, rejects a delayed command from a retired fencing generation, and binds its confirmation evidence to the operation key, generation and reserved physical action.

65. The medium of claim 49, wherein the programs control biological-operations verifier, escrow and commit subsystems whose release records identify actual completed calibration or inspection work; and wherein the instructions preserve the fixed exposure domain and distinguish worst-case reservation from actual spending, require valid instrument and custody evidence before a dependent asset or supply operation, retain a completed-session latch and cumulative upkeep books across requested sessions, and refuse renewed spare capacity without an actually completed, separately paid and qualified reinitialization or separately supplied paid fresh cores, including actual opening state, retained old-state custody and current permissions.

66. A computer-implemented method performed by a governance controller for governing an implicated sub-asset of a finite-register host, comprising: storing a finite policy identifying the sub-asset's actual resource and custody boundary, observable posting and custody ports, preissued denial and retirement authorities with their specified triggering conditions, configured test and response menus, alarm thresholds and finite work reserves; freezing an indexed window of W candidate/evidence positions and authentic associations between those records and actual custody events before selecting a private uniform sample of q distinct positions without replacement, where $1\leq q\leq W$; reserving the complete audit, output and guardian work and verifying each sampled position by an exact predicate satisfied by every complete compliant record in a declared candidate class; computing an actual-invalid-count-indexed table of probabilities of at least one sampled predicate failure and a separately identified sharp guaranteed floor for that event for a certified integer minimum count $1\leq k\leq W$ over the class with at least k invalid positions, and conditioning that floor on the frozen-window, authentic-evidence and completed-q-sample profile; obtaining audit inputs only from the observable records and permitted actual-field reads, without privileged access to the sub-asset's internal program or generating logic; recording an evidence-based alarm, statistical alarm or inconclusive result under the stored policy and charging both correctly and incorrectly imposed response operations under the same event-price menu; upon a qualifying alarm, executing preissued supply-denial and escrow operations, retaining the implicated state, and issuing a bounded-spend certificate only after all declared incoming supply paths and pending inbound commitments are closed or settled and a no-external-supply, no-spent-work-reclamation profile with finite retained stock and authorized conversion yield is verified; selecting containment or whole-module retirement using configured work/capital comparison thresholds and actual qualification checks; when retirement is selected and authorized, retaining the old module and required records, assigning only genuine unexpired assignable rights and a query-selected status record to a supplied pre-authorized successor, and preserving original expiry and the old execution history while checking issuer-authorized current time including elapsed suspension before each assignment or use and inhibiting the operation when that time is unavailable; wherein every digital resource or permission change, its completed-event marker and next-prefix value commit atomically as one durable unit under an immutable event key, recovery returns the stored outcome for a completed key, and any physical command uses a durable one-key execution latch and action-bound confirmation, with replacement or release of affected reservations refused until durable terminal disposition and actuator-boundary fencing establish that no prior command remains executable; and wherein a new session is refused without separately supplied paid qualified cores or an actually completed, conserving, separately paid and qualified reinitialization.

67. The method of claim 66, wherein $W\geq 1$, $1\leq k\leq W$ and $0\leq b\leq W$ are integers, b denotes the actual number of invalid positions in the frozen window, and the controller stores $p(q;b,W)=1-C(W-b,q)/C(W,q)$ as the pointwise exact probability of at least one sampled predicate failure at b and $p(q;k,W)$ as the sharp floor over $b\geq k$, with $C(n,q)=0$ for $q>n$; wherein a minimum-count certificate binds k, the catalogue version, validation predicate and its qualifying premise, a

configured threat-class minimum is not reported as an observed present-window count, and the floor is not reported as the exact power at every $b \geq k$; and wherein the sample is selected independently of the fixed invalid-position placement, the placement and authentic associations remain unchanged through all q comparisons, and a failed evidence or sampling qualification inhibits the completed-sample guarantee.

68. The method of claim 66, wherein the controller enumerates eligible observation counts and threshold tests within the configured finite test menu, reserves configuration, setup, selector, full-comparison, retained-output and qualified guardian work for the selected plan, and maintains actual expenditure separately from its worst-case reservation; wherein a separately configured statistical decision table operates on finite observable transcripts with a stored compliant transcript law $P_0$, and planned false-positive response work is computed as $\Sigma_o\ P_0(o)1_alarm(o)L(o)$, where L(o) is the priced response work for transcript o; and wherein the same observed quarantine or retirement events incur the same actual debits whether their target is compliant or noncompliant, worst-case response stock and work are reserved separately from that expectation, and missed service or expired rights are retained as outcomes rather than erased or counted as recovered credits.

69. The method of claim 66, wherein the cutoff certificate binds an actual cutoff state, each enumerated incoming supply route and the terminal disposition of pending inbound commands, and the controller bounds additional accounted work by R+C, where R is the sub-asset's actual accessible work stock at that state and C is the finite remaining yield of authorized conversion of its retained stock under the verified profile; wherein ordinary unit-fee operations, maintenance and additional fees debit that same allowance without spent-work reclamation; and wherein a new qualified inflow, an unclosed route or a changed conversion or reclamation profile invalidates the old certificate, while the escrow retains the asset's payload, occupancy, unused stock, spent accounts, permissions and history without erasure or renewed capacity.

70. The method of claim 66, wherein the controller compares qualified containment C and retirement R plans at a common paid preparation cut using configured robust capital $E_X=D_\tau+L+A_X+H+2[H-c_X]_+$, for $X \in \{C,R\}$, where $D_\tau$ is common paid detection work, L is common preparation work, $A_X$ is the plan's nonfuture protocol and required live-guardian overhead, H is the remaining finite service horizon and $c_X$ is actual postpreparation spare capacity; wherein this comparator is admitted only with sufficient qualified repair rights and blanks, fixed ordinary and maintenance reserve compartments, the full-trigger at-most-one-binary-posting future profile and actual funding, timing and authority; and wherein a strict retirement-capital preference is selected when $A_R-A_C<2([H-c_C]_+-[H-c_R]_+)$, subject to the stored tie and service-objective policies, without using detection age alone as a retirement rule or charging a parked containment boundary for unrequired live monitoring.

71. The method of claim 66, wherein retirement preserves the whole old module, its custody and required answer paths, requires standing authority ending its old local-answer duties or a qualified retained response path, and maintains cutoff of the implicated incoming route at the successor; wherein the query-selected status record retains each field needed by a fixed receiving-query contract unless the receiver can resolve the identical value from an authorized authoritative versioned record, binds actual asset identity, assigned rights and immutable terms, custody, required provenance and operational prefix, and does not replace actual material or required conditioned-source history with a status code; and wherein an inseparable module containing continuing required records is refused retirement unless a separately authorized and fully priced transfer or retained response path answers those records.

72. The method of claim 66, wherein each actuator-boundary confirmation binds the operation key, fencing generation, reserved source and destination and pre-action and post-action quantity or position evidence; wherein acknowledged cancellation records effective removal or disabling of queued and in-flight commands, delayed or stalled-executor commands from a retired generation are rejected, and missing terminal or fencing evidence preserves an indeterminate lock; wherein permission, calibration and issuer-authorized current-clock validity are checked when actuation is enabled; and wherein a subsequent independent retirement handoff uses a separately supplied, configured, paid and qualified core set and catalogue, while the audit core states, statistics and spent capacity remain retained and no diagnostic flag grants a new assignment edge or refreshes initial spare capacity.

73. The system of claim 37, further comprising a read-only finite audit adapter for a governed sub-asset, an authentic candidate/custody catalogue frozen before private sampling, a supplied selector selecting q distinct positions uniformly without replacement from W positions, and a priced monitor; wherein $1 \le q \le W$, an exact validation predicate is satisfied by every complete compliant record in the declared class, and the monitor stores the actual-count table of probabilities of at least one sampled predicate failure $1 - C(W-b,q)/C(W,q)$ for $0 \le b \le W$, b being the actual number of invalid positions, and the separate sharp floor $1 - C(W-k,q)/C(W,q)$ over $b \ge k$ for a certified minimum $1 \le k \le W$, with $C(n,q)=0$ for $q>n$; wherein the monitor checks authentic associations, fixed placement, selector independence, complete sampling and finite guardian funding before reporting the corresponding guarantee, labels a threat-class minimum separately from a present-window count, prices false-positive response actions under the same configured menu, and retains prior statistics and fresh output records; and wherein the adapter reads observable records and permitted actual fields without access to the governed sub-asset's internal logic and emits no asset-assignment command or new issuer authority.

74. The medium of claim 39, wherein the instructions further store a finite response policy linking observable audit outcomes to preissued supply-denial and escrow conditions, enumerate and confirm closure or settlement of an implicated sub-asset's qualified incoming supply paths, and compute a residual-work bound from its actual retained work stock plus finite authorized conversion yield under a verified no-external-supply and no-spent-work-reclamation profile;

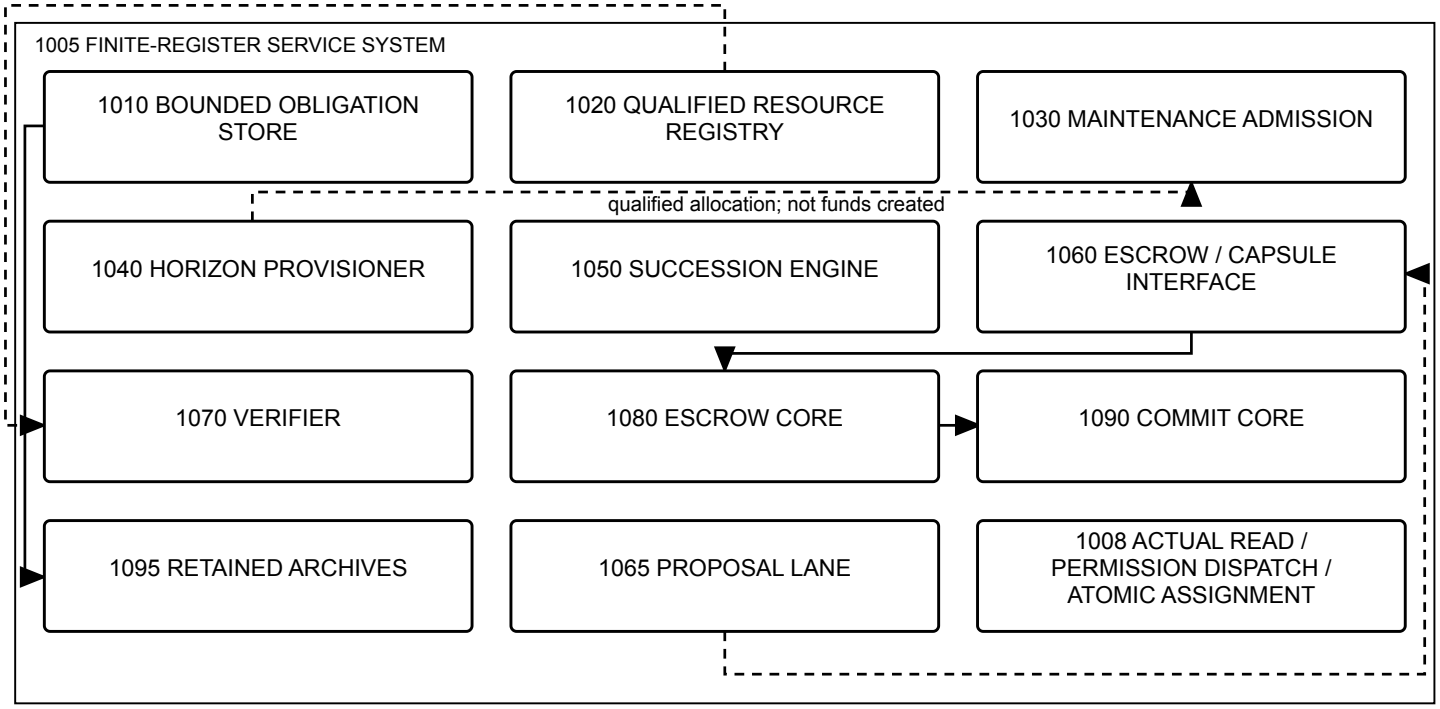
compare containment and whole-module retirement using configured work/capital thresholds, current rights, actual stock and a common comparison cut; and execute an admitted retirement by retaining the old module and required records and assigning genuine unexpired rights with a query-selected status record to a supplied pre-authorized successor without extending expiry; wherein physical execution uses a durable one-key latch, action-bound confirmation and terminal-disposition/fencing gate that refuses replacement and reservation release while an old command remains executable, and a new session requires separately supplied paid qualified cores or actually completed, conserving, separately paid and qualified reinitialization.

75. The method of claim 54, further comprising monitoring authenticated posting, custody, identity and quality-control records of a managed biological sub-asset without privileged access to its internal generating logic; applying a configured evidence or statistical-alarm policy to a contamination or deviation record and pricing a mistakenly imposed hold under the same event menu as a correctly imposed hold; executing preissued supply cutoff and physical-plus-record escrow quarantine while retaining the biological asset, actual stock, provenance and execution state; and selecting the production-role succession of claim 54 over a qualified containment plan using configured work/capital comparators and actual eligibility, expiry and resource checks, with any residual-spend certificate confined to the accounted no-external-supply, no-spent-work-reclamation resource profile; wherein resource changes and completion markers commit atomically under immutable event keys, physical commands use a durable one-key latch and action-bound confirmation, and a replacement command or affected reservation release is refused until durable terminal disposition and actuator-boundary fencing exclude an executable prior command; and wherein subsequent sessions require separately supplied paid qualified cores or actually completed, conserving, separately paid and qualified reinitialization and material is routed to the managed biological process rather than a patient.

## ABSTRACT

A finite-register service system controls maintenance, succession and adversarial sub-assets using qualified resource reservations. An admission controller checks a finite-fuel envelope and delivery, release, custody and recipient conditions, distinguishing funding halt from overflow. A provisioner allocates work and maintenance against effective spare capacity. A succession engine preserves expiry and retained state while assigning rights and query-selected status records to eligible supplied successors. An escrow interface verifies proposals against actual assets and commits guarded assignments. Separately funded guardian machines account for upkeep and paid reinitialization. A read-only audit subsystem computes exact-at-count detection power and certified minimum-count floors over frozen evidence windows and prices false-positive responses. A separate policy controller invokes authorized cutoff and retained-state quarantine. Configured work/capital thresholds select containment or whole-module retirement with lawful rights transfer. Applications include storage, distributed ledgers, biological-asset management and laboratory automation.

**Complete finite-register service architecture**




Description. Show bounded obligation store1010, authoritative qualified-resource registry1020, maintenance admission controller1030, horizon provisioner1040, succession engine1050, escrow/capsule interface1060, verifier1070, escrow core1080, commit core1090 and retained archives1095 within system1005. Show proposal lane1065 entering1060 without entering the asset-transfer path directly.

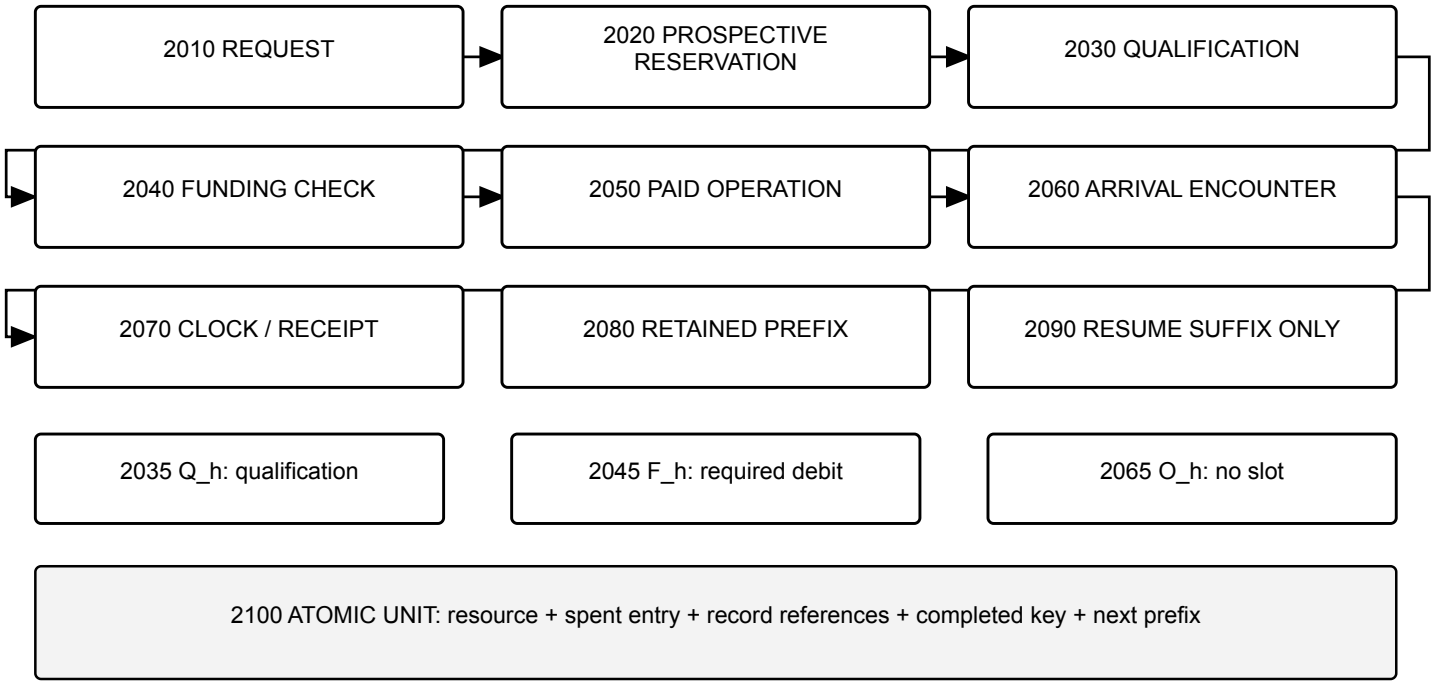
Drawn boundaries. Separate target account1002, interface fees/kits1004 and guardian accounts1006. Put old payloads, actual blank recipients, immutable receipts and spent stores inside the complete roster but outside active service capacity1010. A labeled control-substrate boundary1008 encloses the supplied actual-field-read, permission-confined-dispatch and atomic-assignment primitives. The admitted fault family excludes corruption of those capabilities.

Arrow semantics. Solid arrows identify actual resource/custody transitions; dashed arrows identify proposals or read-only calculations. Mark the return from1040 to1030 “qualified allocation; not funds created.” A dashed read-only reference path from1020 to1070 must not create a second usable asset.

**FIG. 1000 — Sheet 1 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Admission and first-cause state diagram




(1) $H_total + 2D + F \le A_0$; $A_0 = V_0 + (T_0 - T_min)/\varepsilon_W$. (2) $u = \min(\max(0, required - w), accessible_donor, W - w)$. Valid balances: $0 \le w \le W$; invalid balances reject. No actual converter: only actual available $V_0$ funds operation.

States. Request2010 → prospective reservation2020 → stage qualification2030 → funding check2040 → paid operation2050 → arrival encounter2060 → clock/receipt2070. A loop from2070 returns to the next requested word, not to an uncharged fresh opening.

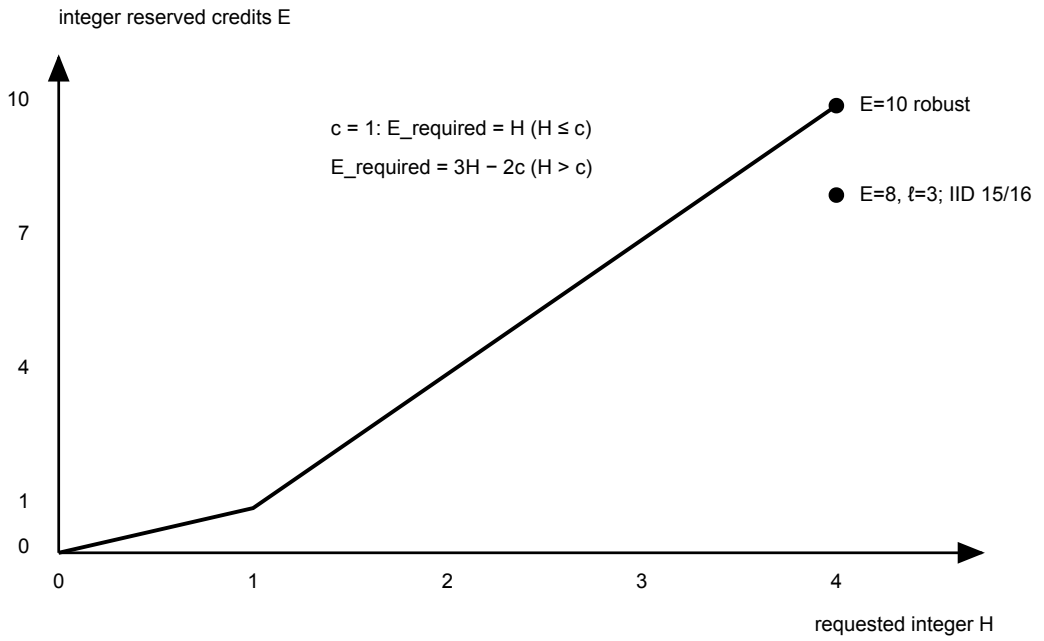
Failure branches. Qualification failure goes to Q_h2035; required debit failure goes to F_h2045; admitted arrival without a slot goes to O_h2065. Show current-prefix store2080 attached to each state, with resume arrow2090 to the outstanding suffix only. At a common fine stage label priority Q_h then F_h; O_h is a later encounter, not an interchangeable label.

Inset. Display equation(1), valid top-up equation(2), and an optional-maintenance rejection branch that may continue to encounter. Add atomic update box2100 containing the resource change, spent entry, retained-record references, durable completed-event marker and next prefix under one immutable event key. Crash/recovery exposes the whole pre-update or whole post-update state; an existing completed key returns its stored result without a second mutation. External nontransactional effects are outside this box. Include the empty-register/no-ordinary-credit example to distinguish funding halt from overflow. A retained debit does not roll back on interruption.

**FIG. 2000 — Sheet 2 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Horizon provisioning frontier chart




3010 QUALIFICATION MASK: wrong ID / deadline / route / headroom → no robust certificate

Axes. Horizontal requested integer H; vertical integer reserved credits E. For a declared c≥0, plot E_required=H for H≤c and E_required=3H−2c for H>c, joining at H=c. Label the chart with its configured profile and admission domain.

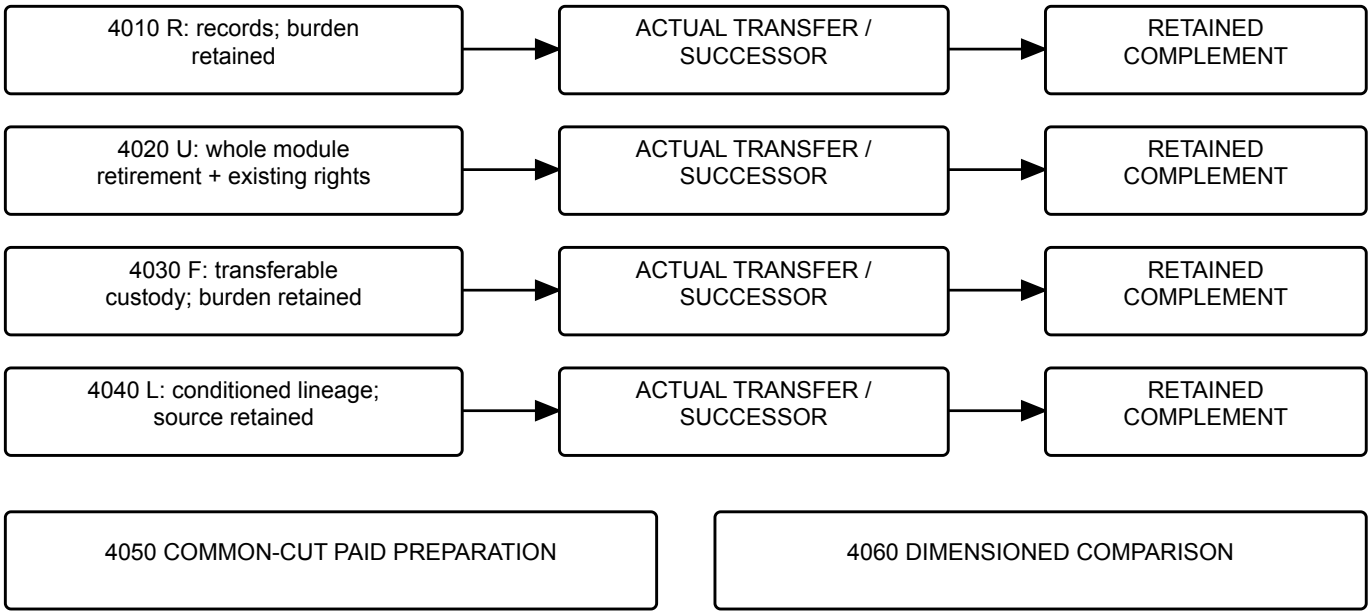
Allocation inset. E→H ordinary credits +2m maintenance credits + odd leftover. Print the admission domain B≥1, integer B/z/H, 0≤z≤B, H≥0, at most one binary arrival per funded word, full-trigger maintenance before the unrevealed arrival and an accessible two-credit local service purse; robust all-word counting is not conditioned on IID probabilities. Show c+m as virtual spare capacity, with m capped by actual blanks, releases and opportunities. A separate qualification mask3010 marks “no robust certificate” where resources have wrong IDs, wrong deadlines, unavailable routes or missing headroom, even above the line.

Worked markers. At c=1,H=4 mark E=10 robust; mark E=8 with ℓ=3 and IID(a=1/2) survival 15/16. Mark the missing-release E=10 instance as unqualified, not as lying on the positive frontier. Optional survival contours must be labeled fixed horizon/fixed allocation/IID; no cross-horizon hazard is drawn.

**FIG. 3000 — Sheet 3 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Succession modes and dominance map




Layout. Four rows R4010, U4020, F4030 and L4040. Each has old module, actual transferred objects, successor and retained complement. Label R “records, burden retained”; U “whole-module retirement + existing rights”; F “specified transferable custody, burden retained”; L “actual conditioned lineage/source retained.” Draw no arrow that converts old payload erasure into fresh capital.

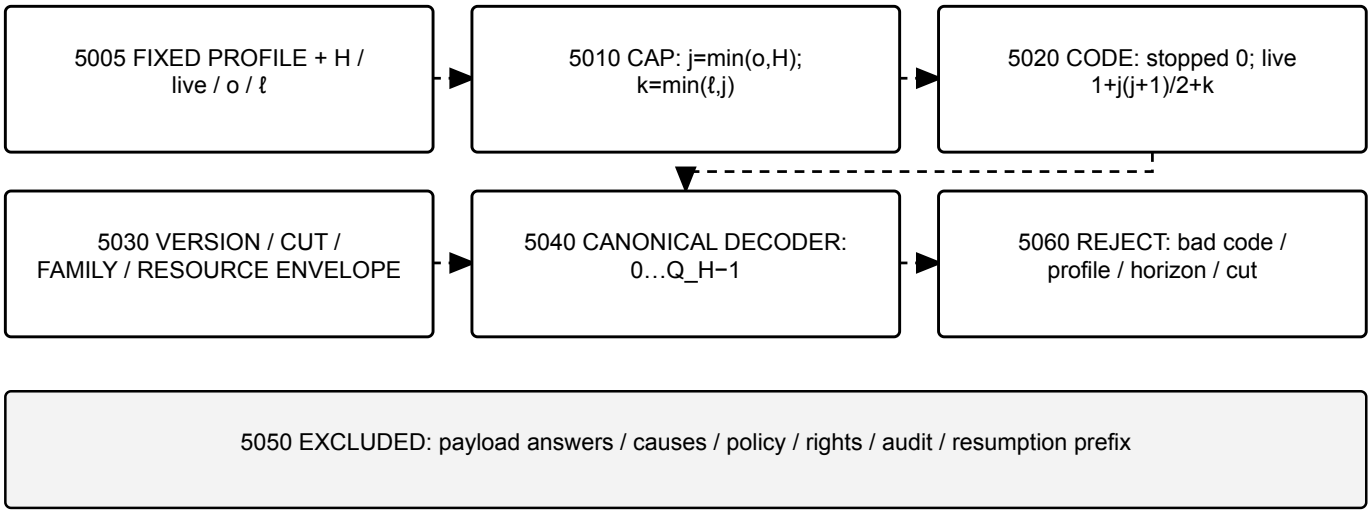
Comparison panels. A common-cut timeline4050 shows both branches' paid preparation. A result panel4060 has separate columns: role survival, actual work, reserved capital, elapsed words, residual stock and old-root status. Show each comparison within its stated dimension.

Exact annotations. For error-free U with the specified base menu, show 5 versus repair6/8 under the same two-module roster and role contract. Label this “base transfer menu; interface and guardian accounts separate.” Alongside, show the integrated seven-slot table48/63/78 versus24/25/26 for b=0/1/2, labeled with its complete configuration menu. A no-discount branch carries equation(6); an expiring-rights branch labels early exercise with unchanged absolute expiry.

**FIG. 4000 — Sheet 4 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Restricted status snapshot and excluded queries




Encoder. Inputs H, live, actual residual o,ℓ and fixed profile/family5005 feed cap block5010. The profile requires independent Bernoulli(a) future arrivals with fixed 0<a<1, fixed full-trigger policy, status-only observations and no additional conditioned-source variable. Cap rule: j=min(o,H), k=min(ℓ,j). Code block5020 maps stopped→0 and live→1+j(j+1)/2+k. Envelope5030 retains version, H, fixed a, advanced cut, family and future-resource identifiers. Decoder5040 checks the declared profile, integer horizon, alphabet and exact cell count and requires integer code 0…Q_H−1 before selecting a triangular interval. Reject branch5060 handles unused or noncanonical words and profile/horizon/cut mismatches without constructing live state or permission. At H=2 there are seven valid labels0…6; binary three-bit code 7 rejects.

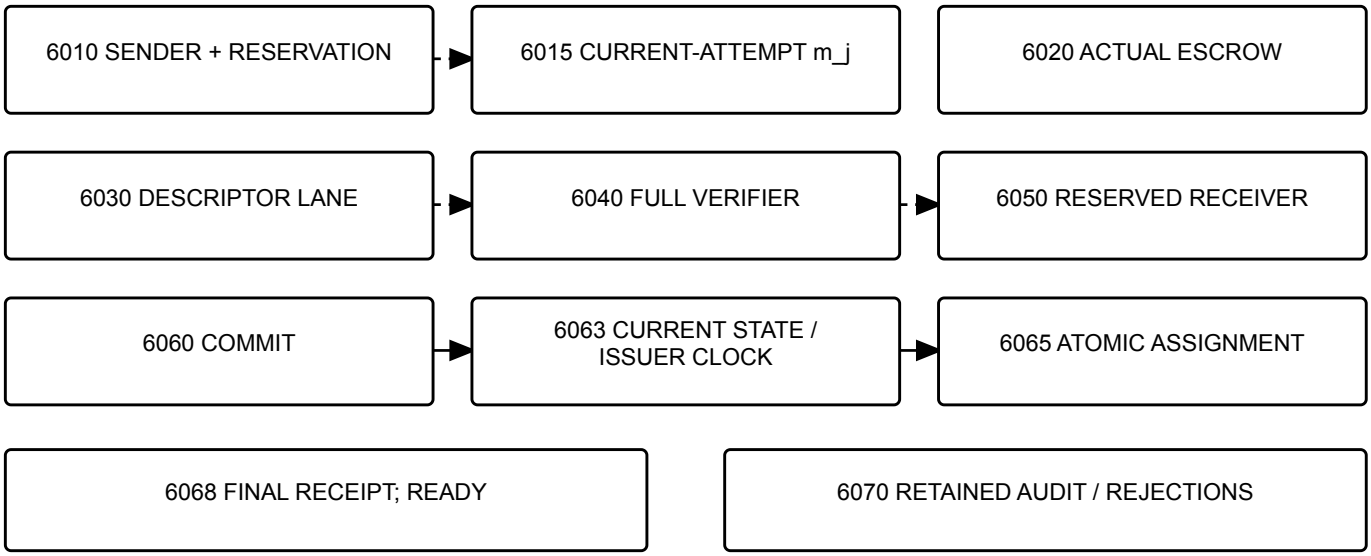
Small instance. H=1 diagram has stopped:(0,0); live j=0:(1,0); live(1,0):(1,1/2); live(1,1):(1,1), where entries are current-live and alive-next for a=1/2. Label $Q_1$=4 deterministic classes; linear rank 3; no “three bits” label.

Boundary. Excluded/read-separate panel5050 lists pending payload answers, cause labels, arbitrary policy choices, rights/permissions, audit history and resumption prefix. Dashed edge indicates that adding one of these queries requires a refined record. Distinct actual assets may already convey the class; do not mandate a duplicate synopsis.

**FIG. 5000 — Sheet 5 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Escrow/capsule protocol sequence




Lifelines. Sender6010, actual escrow6020, fallible descriptor lane6030, verifier6040, reserved receiver6050, commit6060 and audit6070. Begin with actual asset reservation and authoritative session-manifest generation. Expected-descriptor block6015 inserts current protected slot nonce j into the retained session fields to form m_j; verifier6040 compares all received fields to that current-attempt tuple, not a fixed nonce1 tuple. Retry fees do not reduce promised purse v_p. The only object sent through6030 is a descriptor.

Per-slot sequence. Fresh kit/prefix → ordinary 1 and d debits → complete encode/carrier → full compare → actual terms/lock/balance/receiver checks → acceptance or retained rejection. Show a rejected nonce and a subsequent valid nonce. After acceptance draw paid IDLE slots until the fixed final cut. The first current duplicate may be valid; later replay does not duplicate assignment.

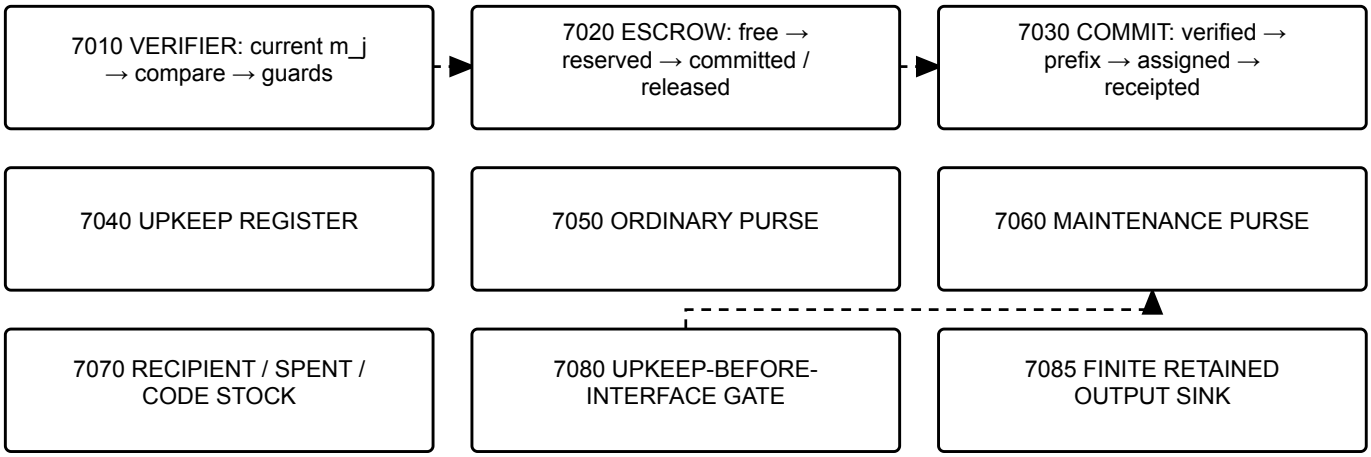
Commit. Current-state and issuer-authorized current-clock check6063 precedes atomic assignment6065; final receipt6068 follows. The clock-domain label identifies issuer time or authoritative epoch, never a paused private counter. An unavailable clock inhibits; delayed recovery outside the envelope invalidates the prior completion/robust certificate. Each mutation, its completed-event marker and next prefix are one durable atomic update; assignment6065 includes its assigned marker, while receipt6068 has a later key. Mark a halt between them “assets assigned; READY=false; resume receipt suffix.” On no valid proposal by N, abort keeps actual assets and old access only under original still-valid terms. No fee refund arrow.

Price panel. $d=(e+k_c+v)t+g$; verification $vt+g$; configuration $K_0+\kappa N$; common-cut differential $b(d+\kappa)$. State that kits/factors are supplied and $N=b+1$. A four-cell obstruction strip lists exclusive-asset loss, refusal, authority/descriptor inconsistency and absolute expiry. Put corrupted sole authority outside the positive trust boundary; do not show a perfect detector for it.

**FIG. 6000 — Sheet 6 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

**Three budgeted guardian machines**




Verifier7010. Retained authoritative session manifest + current protected slot nonce → expected complete m_j → full-cell comparison/actual-state guard tests → accepted or rejected; processed-session latch retained. The declared finite program/catalog, pointer recipients and scanner menu include this bounded per-slot derivation, with inter-session reference refresh admitted separately under FIG. 8000. Escrow7020. Free → reserved → committed, or reserved → released subject to original terms. Commit7030. Unverified → verified → transfer-prefix → assigned → receipted; rejected/aborted and halted-prefix alternatives remain visible.

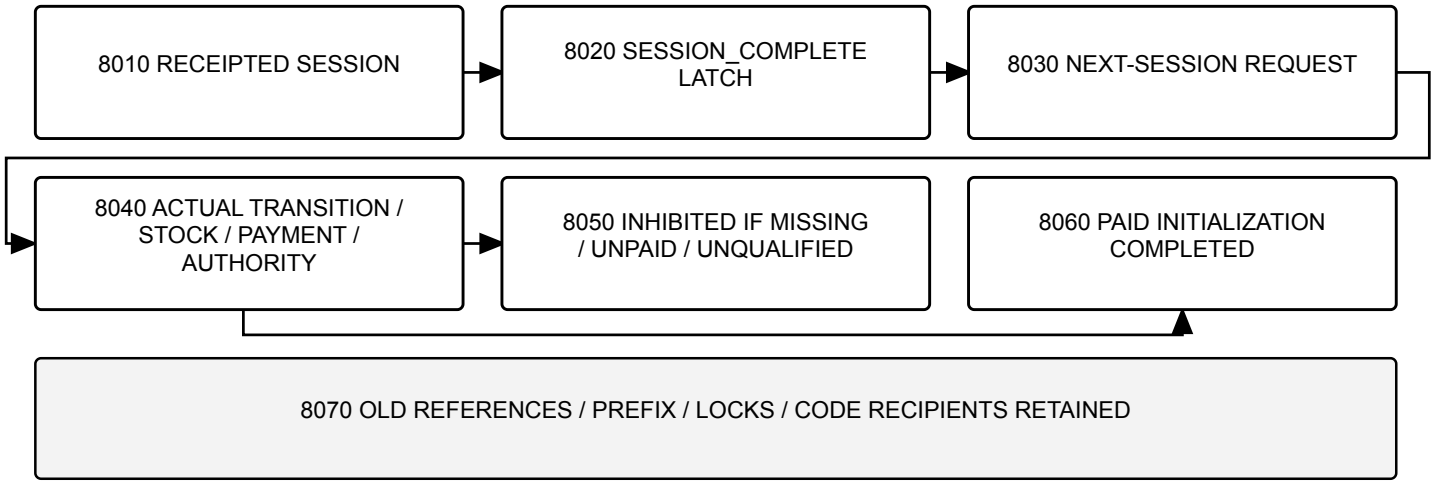
Under each core. Separate upkeep register7040, ordinary purse7050, maintenance purse7060 and recipient/spent/code stock7070; label worst-case reservation $E_i=T+2[T-c_i]_+$, integer $1\le c_i\le B_i$, at most one binary posting per funded upkeep word, and full-trigger upkeep before the unrevealed posting. Keep actual pathwise spending separate from this robust reserve. These cannot draw upkeep from the locked beneficiary purse. Show upkeep-before-interface gating7080 and distinct core-F_h/core-O_h/core-Q_h alarms.

Setup/pricing inset. Three one-cell setup epochs, J=9; general serial $J=3(t+2)$. Every fine-stage resource mutation, completed-event marker and next prefix is one keyed durable unit; the commit guard includes actual issuer-clock expiry. A changed catalog, pointer count or timing outside the menu requires repricing and requalification. Display total $M=3T+2\Sigma_i[T-c_i]_+$ as distinct from scanner fees. Mark FC-1 and atomic assignment as supplied control-substrate primitives with their specified fault domain. A finite sink7085 retains actual outputs and headroom; its arrows denote retained-state custody.

**FIG. 7000 — Sheet 7 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Session re-entry interlock and cumulative account




Main path. Receipted session8010 → SESSION_COMPLETE latch8020 → next-session request8030 → transition/stock/payment/authority validator8040. Missing connector, unpaid transition or unqualified stock goes to INHIBITED8050. An affirmative path8060 is labeled “only after separately specified and actually completed paid initialization.” Validator8040 compares authoritative actual opening state, retained old-state custody, current permissions, fresh references and exposure envelope; a document alone is not a transition. Old reference, prefix, locks and code recipients remain8070. The validator admits the supplied initialization; the diagram adds no connector operation.

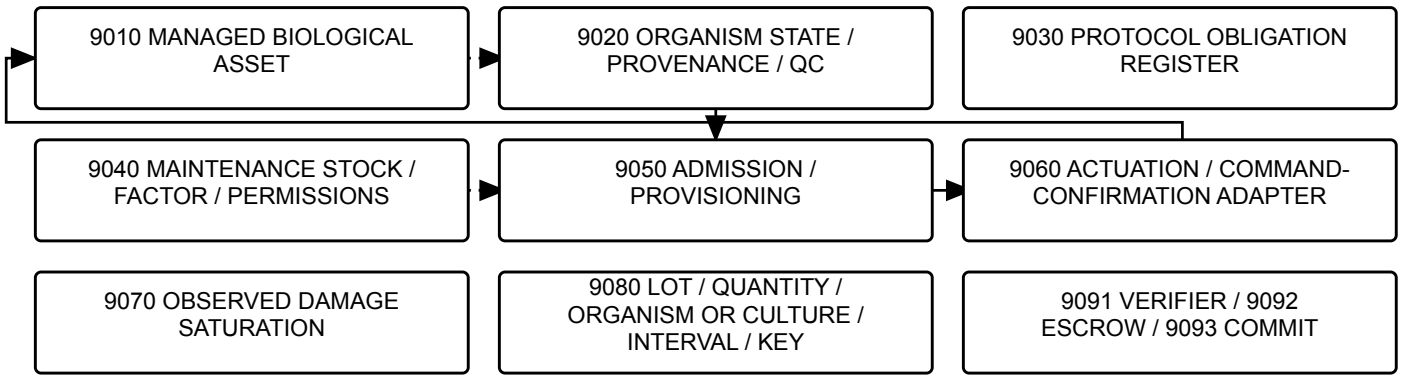
Accounting inset. The same retained upkeep books across an n-session horizon require worst-case reservation $M(nT;c)$ instead of $nM(T;c)$, within the same qualified at-most-one-binary-posting profile. Actual expenditure is recorded separately; an all-IDLE path need not consume every maintenance pair. Fresh cores enter as new counted inventory with paid setup. Retain $W_{link}$ as the separately specified connection price in equation(16); absence of a priced connection inhibits execution.

Comparison output. For the disclosed numerical configuration show the zero-connection-cost comparison ceilings $-40,-75,-110$ at $n=1,2,3$. Label them “connection cost excluded from comparison bound; execution requires paid initialization.” Positive connection charges reduce these values. Draw the actual-state/payment admission gate on every next-session path.

**FIG. 8000 — Sheet 8 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

**Bioprocess admission, factor delivery and guardian control**




Architecture. Show biological asset9010, passage/generation and QC observation record9020, bounded protocol-obligation register9030, actual maintenance-stock/factor and permission registry9040, admission/provisioning controller9050 and journaled intervention/dispense adapter9060. The managed asset comprises one or more organisms, a culture, a population or preserved biological material. In embodiments, it is an ex vivo culture, bioreactor culture or managed breeding stock. The asset-records system and actuation interface supply the record and command paths; LIMS, bioreactor controllers and laboratory-automation adapters are embodiments. Show actual inventories separately from credit and permission fields.

Control flow. Observation → current-time, identity/QC and supply checks → finite-fuel and fixed-profile reservation → guarded intervention authorization → physical confirmation → addressed discharge/receipt. Preserve passage history and actual QC evidence. Draw distinct F_h funding, Q_h qualification and O_h register-overflow branches and a separate observed damage-saturation indicator9070. B,z,H and the frontier label refer to the controller-obligation profile, not biological division rates.

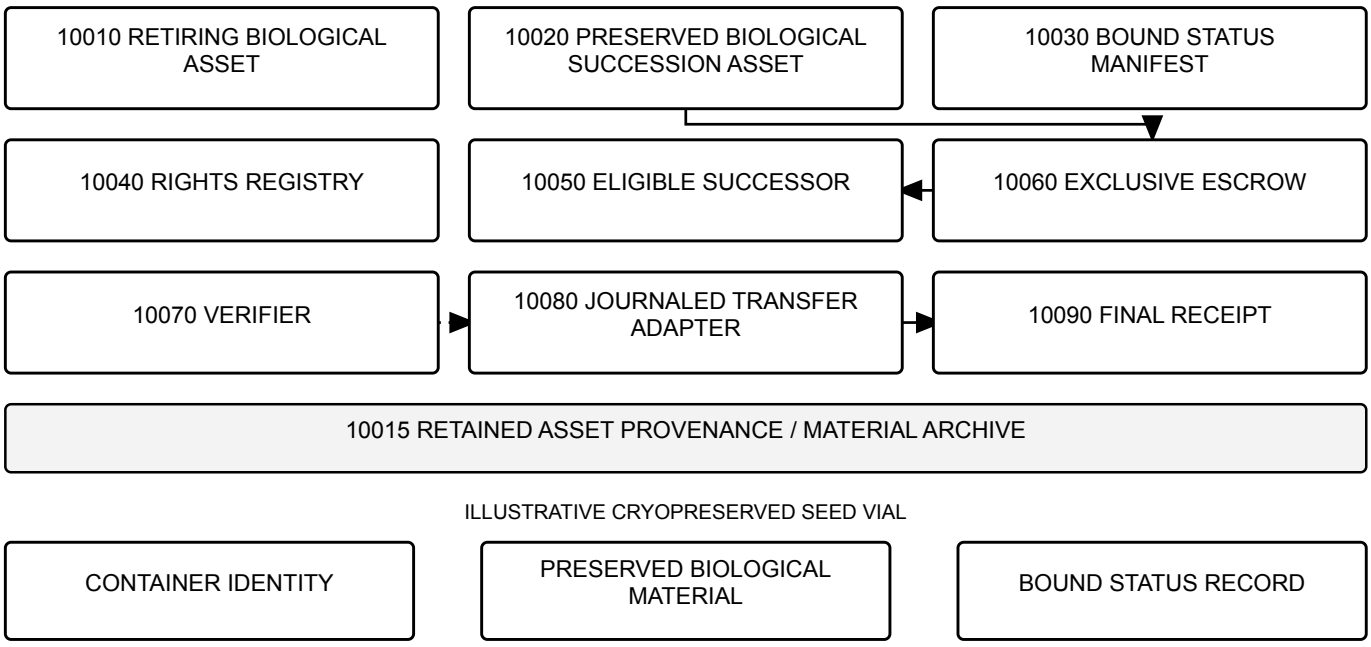
Factor-credit inset. An issued credit9080 binds actual lot, maximum dispense quantity, controlled organism or culture, valid interval and immutable operation key. The organism-state observation interface supplies the actual asset observations; culture-named record fields retain the E10.5 mappings. The dispenser gate reads y, its observation uncertainty, y_max and the declared absolute finite response interval; it compares the requested-delivery upper increment U plus residual-response bound R with the available headroom of y in common process-variable units. Missing or invalid profile bounds or observations inhibit admission. Show retained allowances for delivered and still-authorized material, with no release until actual terminal disposition and residual response are resolved. Credit exhaustion inhibits a new supply command; a separate observation path continues to report culture state. Reproduction and label changes have no credit-issuance arrow. Show the durable EXECUTION_STARTED latch, later physical-confirmation state, and INDETERMINATE lock/reconciliation branch; repeated keys do not re-actuate. A replacement-key branch requires a durable completed-action or effective-cancellation disposition with no executable old command remaining, and actuator-boundary fencing of the old generation. The boundary rejects delayed and stalled-executor commands from a retired generation. Missing terminal/fencing evidence preserves INDETERMINATE and all affected reservations. Bind confirmation to key, generation, reserved source/destination and pre/post quantity or position; gate actual actuation on current permission, calibration and command-validity interval. Re-entry passes through the paid initialization validator of FIG. 8000.

Guardian inset. Verifier9091, escrow9092 and commit9093 each have separate upkeep reserves, actual calibration releases and retained books. Label each qualified worst-case reserve $T+2[T-c_i]_+$ with the one-binary-posting, pre-arrival upkeep and fixed-compartment domain. Current calibration and clock validity gate the downstream command. Retained-core repeated sessions use cumulative upkeep. Calendar-time environmental support is a separate reserved account. In embodiments, these automation subsystems serve production, biobank, agricultural or environmental-management facilities. Show biological-asset locations and supply lots coupled through the command-and-confirmation adapter; laboratory automation is an embodiment.

**FIG. 9000 — Sheet 9 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Biological succession and biobank custody sequence




Lifelines. Retiring biological asset10010, actual preserved biological succession asset10020, bound status manifest10030, rights registry10040, pre-authorized successor10050, exclusive escrow10060, verifier10070, journaled physical-transfer adapter10080 and final receipt10090. Retain old asset provenance history and material in a named archive10015; biological-line and passage-history records are embodiments. A custody arrow moves the actual unique material; a dashed descriptor arrow carries no asset authority. Asset10020 comprises a whole organism, a culture, a biological line, a propagule, or preserved biological material sufficient to establish a successor organism, culture or line. A cryopreserved seed vial is the illustrated material embodiment; show its container identity separately from the material and bound status record.

Snapshot inset. The actual succession asset plus task-selected status record form the minimal-transfer snapshot. Show fixed receiving queries selecting necessary identity, provenance-history reference, QC/preservation evidence, quantity, rights and prefix fields. A field can be omitted only if identical data are already uniquely available in an authorized authoritative record. Compare actual succession-asset identity and current custody/preservation evidence with the versioned receiving-query record before binding the class code; a failed or unavailable comparison inhibits succession-readiness. Keep material, biological lineage and permission records distinct from the optional E5 controller-status code. Label that code with its fixed independent Bernoulli(a), full-trigger, status-only/no-feedback/no-additional-conditioned-source domain; it does not replace the actual succession asset.

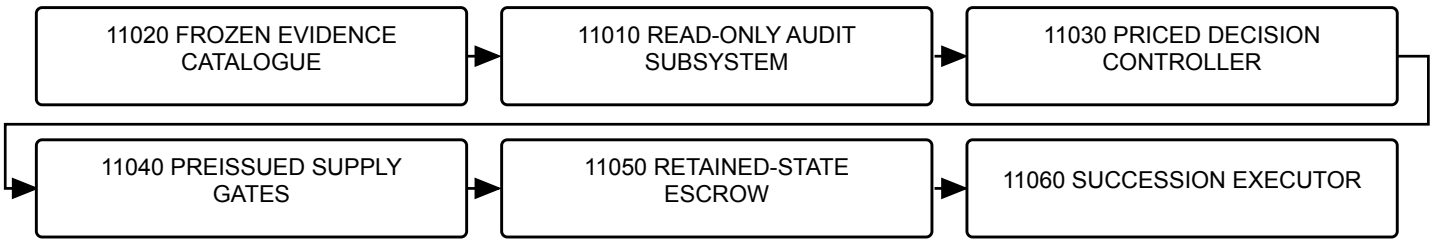
Succession sequence. Check preissued retirement and successor eligibility → reserve the succession asset, receiving resources and establishment cost → preserve original assignability, uses, consent/revocation flags and absolute expiry → perform paid establishment → confirm actual initialized successor and retained old custody → final readiness. Every expiry-sensitive action checks the issuer-approved current clock including suspension. Missed expiry inhibits the outstanding action; a completed assignment remains recorded.

Escrow sequence. Derive current-attempt m_j → reserve assay aliquots and full verification fee → evaluate identity/QC and actual custody → accept or retain rejection → paid scheduled padding → current-state recheck → authorized physical action → confirmed receiving custody → final receipt. A separate strip identifies loss, refusal/QC-fail, identity/authority inconsistency and expiry. Material QC failure holds quarantine independently of descriptor retries. Show per-key EXECUTION_STARTED, CONFIRMED and INDETERMINATE states and the reconciliation lock; an accounting debit alone is not a sample-move receipt. Permit replacement and reservation release only after durable completion or effective cancellation with no executable old command remaining and retirement of the old fencing generation at the actuator boundary. Reject delayed or stalled-executor commands from that generation; absent terminal or fencing evidence retains the lock. Bind confirmation to key, generation, reserved source/destination and pre/post quantity or position. Check current permission, calibration and command-validity interval at actuation; retain completed actions whose confirmations arrive later.

**FIG. 10000 — Sheet 10 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.

## Adversarial sub-asset governance and retained-state quarantine




Certificate event: at least one failed complete sampled comparison

Exact at actual b: $p(q;b,W)$ | Sharp certified class floor: $p(q;k,W)$, $b \geq k$

INVALID_EVIDENCE / STATISTICAL_ALARM / NO_SAMPLED_FAILURE / INCONCLUSIVE

ALL ROUTES + PENDING COMMANDS SETTLED: accounted additional work ≤ R + C

G4: $E_X = D_\tau + L + A_X + H + 2[H - c_X]_+$

G5: $A_R - A_C < 2([H - c_C]_+ - [H - c_R]_+)$

QUALIFIED CONTAINMENT / RETIREMENT | TIE POLICY | UNAVAILABLE

Architecture. Show read-only audit subsystem11010, frozen evidence catalogue11020, priced decision controller11030, preissued supply gates11040, retained-state escrow11050 and succession executor11060. Actual sub-asset/module identifiers, unique custody edges, authentic before/after associations, current rights and issuer clocks are separate from questioned proposals. No arrow supplies the governed sub-asset's internal program, hidden selector or generating logic to the audit adapter.

Audit flow. Freeze W positions and actual associations → reserve q complete inspections, selector/output factors and guardian upkeep → select private uniform distinct positions after placement freezes → exact record checks and retained statistics → INVALID_EVIDENCE / STATISTICAL_ALARM / NO_SAMPLED_FAILURE / INCONCLUSIVE. Show separate labels $p(q;b,W)$ for the exact probability of at least one failed complete sampled comparison at actual b and $p(q;k,W)$ for that event's sharp class floor over $b \geq k$. Evaluate response thresholds and statistical-alarm rules separately; their outputs do not acquire the certificate-hit probability. A configured threat-class minimum is not an observed current-window count. Missing or changed evidence, incomplete q, changed placement or invalid selector qualification inhibits the corresponding guarantee.

Pricing inset. The configured test menu includes full setup, configuration, reads, complete comparisons, output and upkeep. The false-positive branch uses the same observed-event fee path as the correct-alarm branch, plus any actually executed hold/retirement. Display expected wrong-hold work separately from reserved maximum response capital and actual expenditure. A parked protected lock has no invented perpetual live-duty bill; scheduled responsive HOLD has its explicit funded path.

Containment sequence. Qualifying alarm → preissued denial-authority and triggering-condition check → close every declared incoming route → settle or effectively cancel/fence pending inbound commands → reconcile R and finite conversion yield C → certify additional accounted work≤R+C within the no-external-supply/no-reclamation profile → retain whole state and custody. An unclosed route or changed profile inhibits the old certificate. Show INDETERMINATE with no replacement/reservation release until terminal disposition and actuator-boundary fencing exclude old executable commands.

Selection and succession. Compare qualified containment and whole-module retirement at a common cut using configured service/work/capital thresholds. Show $E_X$ and the strict-capital inequality G5, a tie-policy branch and an UNAVAILABLE branch for missing authority, funds, rights or answer paths. A retirement branch checks pre-authorized successor, actual supplied stock, immutable expiry at the current clock and continuing-query obligations, retains the old module, and transfers genuine rights plus a query-selected status record. Cutoff also guards the successor route. A later independent handoff uses separately supplied paid qualified cores and catalogue; no flag-to-free-reset arrow is permitted.

Applications. Insets depict a permissioned-ledger partition, storage-fleet module and managed biological sub-asset. Biological quarantine is physical-plus-record custody through the E10.1 adapter; excision follows the B2 production-role path. All physical actions retain key/generation/source/destination/quantity-or-position confirmation and current permission/calibration gates. The resource bound is not a biological-growth or viability prediction.

**FIG. 11000 — Sheet 11 of 11**

Solid: actual custody/resource or named state progression. Dashed: descriptor/read-only reference.